

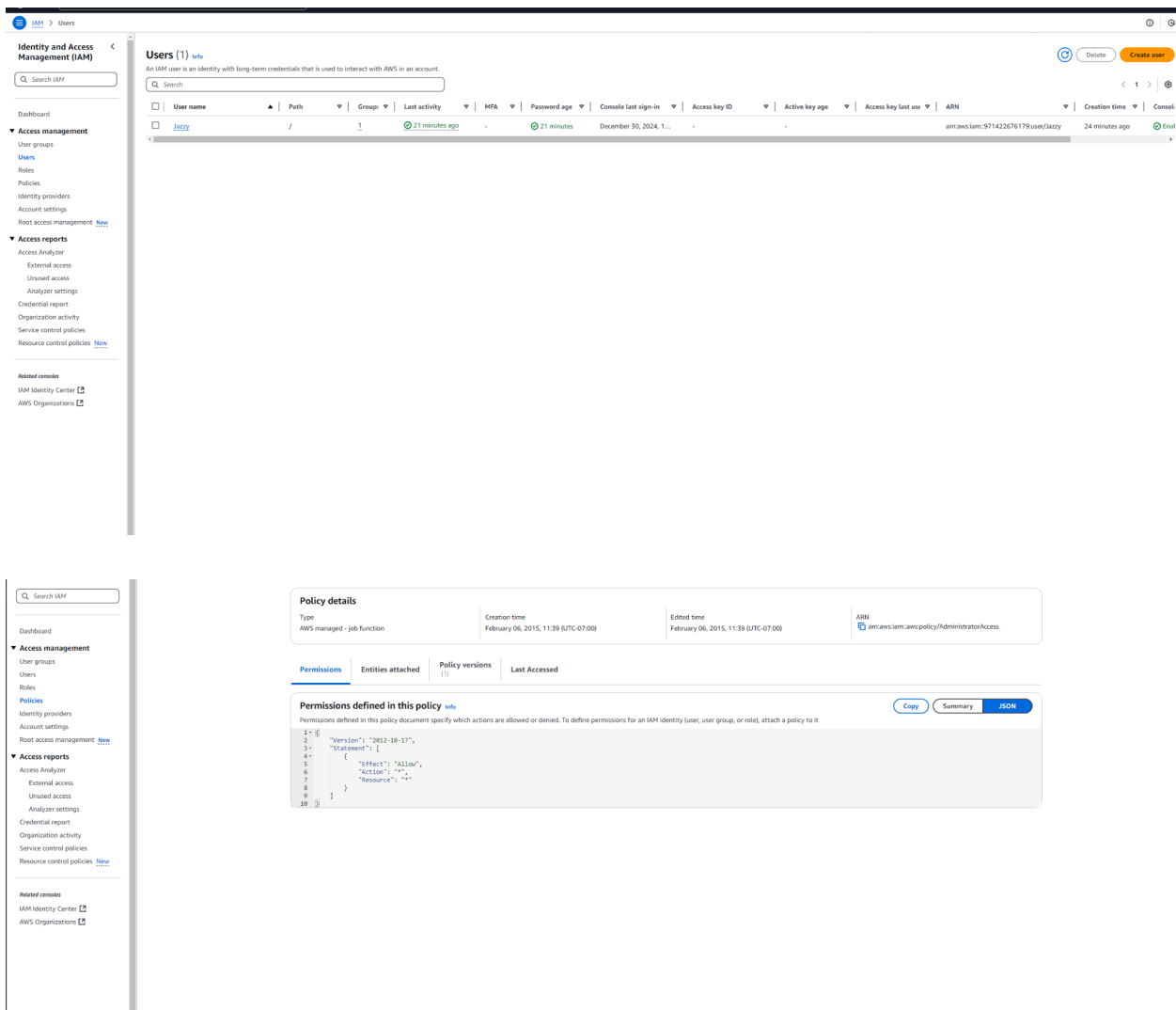
My main goal is to study for AWS SAA and become familiar with it's related services.

Day 1:

IAM

Hands-on lab creating an IAM user.

I created a user named Jazzy and played around a bit with groups and permissions



Hands-On Lab Creating a password policy in IAM

At my real job, I have created and implemented a password policy. I have it match some NIST standards of: 8 minimum characters, at least one upper and lower case letter, at least one number and one symbol. I have a password re-use set to 10 passwords as well

Edit password policy [Info](#)

Password policy

☐ IAM default
Apply default password requirements.

☒ Custom
Apply customized password requirements.

Password minimum length.
Enforce a minimum length of characters.

characters
Needs to be between 6 and 128.

Password strength

☒ Require at least one uppercase letter from the Latin alphabet (A-Z)
☒ Require at least one lowercase letter from the Latin alphabet (a-z)
☒ Require at least one number
☒ Require at least one non-alphanumeric character (!@# \$ % ^ & * () _ + - = [] | ' '')

Other requirements

☐ Turn on password expiration
☐ Password expiration requires administrator reset
☐ Allow users to change their own password
☒ Prevent password reuse
Remember password(s)
Needs to be between 1 and 24.

[Cancel](#) [Save changes](#)

I assigned MFA to my root account. Having access to the root account is very dangerous if it gets compromised

IAM Dashboard [Info](#)

Security recommendations 

 **Add MFA for root user**
Add MFA for root user - Enable multi-factor authentication (MFA) for the root user to improve security for this account.

 **Root user has no active access keys**
Using access keys attached to an IAM user instead of the root user improves security.

[Add MFA](#)

Select MFA device [Info](#)

MFA device name

Device name

This name will be used within the identifying ARN for this device.

Maximum 64 characters. Use alphanumeric and '+', '.', '@', '-', '_' characters.

MFA device

Device options

In addition to username and password, you will use this device to authenticate into your account.



Passkey or security key

Authenticate using your fingerprint, face, or screen lock. Create a passkey on this device or use another device, like a FIDO2 security key.



Authenticator app

Authenticate using a code generated by an app installed on your mobile device or computer.



Hardware TOTP token

Authenticate using a code generated by Hardware TOTP token or other hardware devices.

[Cancel](#)[Next](#)

MFA is set:

Try the new sign in UI
See our new improved Amazon Web Services sign in experience before we officially launch.

Enable new sign in



Multi-factor authentication

Your account is secured using multi-factor authentication (MFA). To finish signing in, turn on or view your MFA device and type the authentication code below.

Email address: camdengowans77@gmail.com

MFA code

Submit

[Troubleshoot MFA](#)

[Cancel](#)



© 2024, Amazon Web Services, Inc. or its affiliates. All rights reserved.

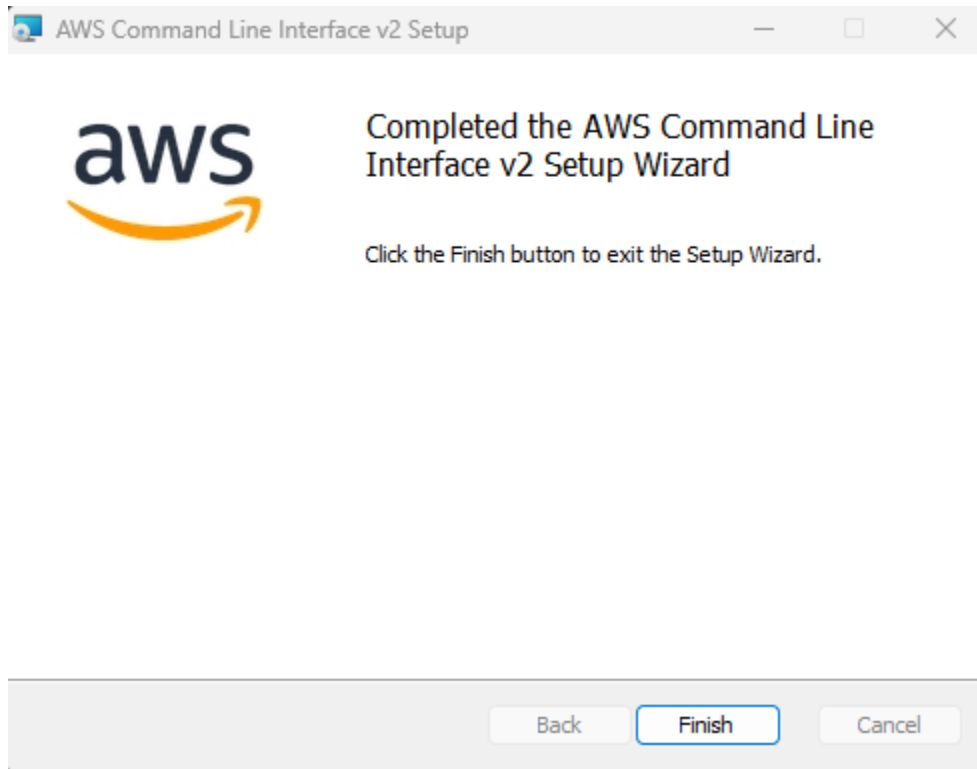
English

IAM Dashboard [Info](#)

Security recommendations 0

- ✓ Root user has MFA
Having multi-factor authentication (MFA) for the root user improves security for this account.
- ✓ Root user has no active access keys
Using access keys attached to an IAM user instead of the root user improves security.

I wanted to venture toward AWS CLI, so I installed it on my desktop|



It is installed now

A screenshot of a Windows Command Prompt window. The title bar says 'Command Prompt'. The text inside the window shows the system information: 'Microsoft Windows [Version 10.0.26100.2605]' and '(c) Microsoft Corporation. All rights reserved.' The user has entered the command 'aws --version' at the prompt 'C:\Users\black>'. The output of the command is 'aws-cli/2.22.26 Python/3.12.6 Windows/x64 exe/AMD64'. The prompt 'C:\Users\black>' is shown again on the next line, indicating the command has been executed successfully.

Creating Access Keys in AWS IAM

This is for AWS CLI

Access key best practices & alternatives [Info](#)

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

- ☒ **Command Line Interface (CLI)**
You plan to use this access key to enable the AWS CLI to access your AWS account.
- ☐ **Local code**
You plan to use this access key to enable application code in a local development environment to access your AWS account.
- ☐ **Application running on an AWS compute service**
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.
- ☐ **Third-party service**
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.
- ☐ **Application running outside AWS**
You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.
- ☐ **Other**
Your use case is not listed here.

Alternatives recommended

- Use [AWS CloudShell](#), a browser-based CLI, to run commands. [Learn more](#)
- Use the [AWS CLI V2](#) and enable authentication through a user in IAM Identity Center. [Learn more](#)

Confirmation

- ☒ I understand the above recommendation and want to proceed to create an access key.

[Cancel](#)

[Next](#)

Access keys should be blurred out

Access key created

This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

Step 1

Access key best practices & alternatives

Step 2 - optional

Set description tag

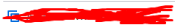
Step 3

Retrieve access keys

Retrieve access keys [info](#)

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
	 Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

[Download .csv file](#) [Done](#)

This is now configured correctly through CMD

```
Command Prompt
Microsoft Windows [Version 10.0.26100.2605]
(c) Microsoft Corporation. All rights reserved.

C:\Users\black>aws --version
aws-cli/2.22.26 Python/3.12.6 Windows/11 exe/AMD64

C:\Users\black>aws configure
AWS Access Key ID [None]: 

C:\Users\black>aws configure
AWS Access Key ID [None]: 
AWS Secret Access Key [None]: 
Default region name [None]: us-west-1
Default output format [None]:

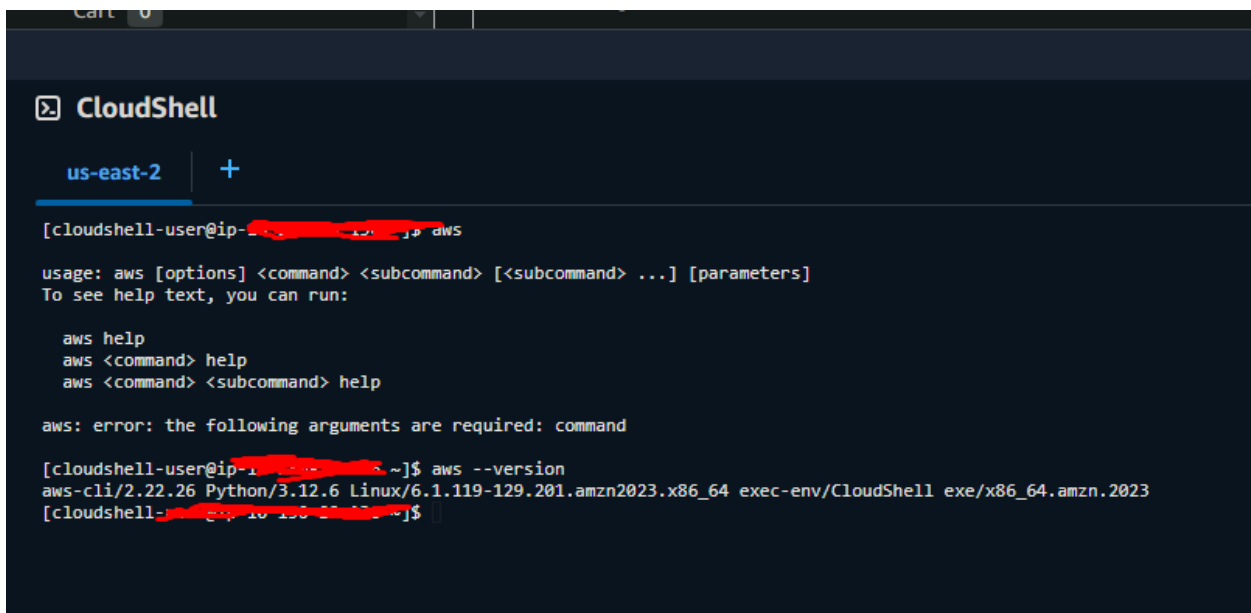
C:\Users\black>
```

I verified this by listing IAM users: `aws iam list-users`

```
C:\Users\black>aws iam list-users
{
  "Users": [
    {
      "Path": "/",
      "UserName": "Jazzy",
      "UserId": "AIDAJAZZY",
      "Arn": "arn:aws:iam::123456789012:user/Jazzy",
      "CreateDate": "2024-12-30T19:13:20+00:00",
      "PasswordLastUsed": "2024-12-30T19:16:05+00:00"
    }
  ]
}
```

Hands-on Lab Exploring AWS CloudShell

I wanted to explore AWS CloudShell. It's basically a web version of AWS command line.



The screenshot shows the AWS CloudShell interface. At the top, there's a tab labeled "us-east-2" with a plus sign to its right. Below the tab, the prompt is "[cloudshell-user@ip-10-100-20-100 ~]\$". The user has entered the command "aws". The interface displays the usage for the "aws" command: "usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]". It also provides instructions on how to see help text: "To see help text, you can run: aws help", "aws <command> help", and "aws <command> <subcommand> help". Below this, an error message is shown: "aws: error: the following arguments are required: command". Finally, the user enters "aws --version", and the output is displayed: "aws-cli/2.22.26 Python/3.12.6 Linux/6.1.119-129.201.amzn2023.x86_64 exec-env/CloudShell exe/x86_64.amzn.2023".

```
CloudShell
us-east-2 +
[cloudshell-user@ip-10-100-20-100 ~]$ aws
usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:
    aws help
    aws <command> help
    aws <command> <subcommand> help
aws: error: the following arguments are required: command
[cloudshell-user@ip-10-100-20-100 ~]$ aws --version
aws-cli/2.22.26 Python/3.12.6 Linux/6.1.119-129.201.amzn2023.x86_64 exec-env/CloudShell exe/x86_64.amzn.2023
[cloudshell-user@ip-10-100-20-100 ~]$
```

Listing AWS IAM users

```
[cloudshell-user@ip-10-10-10-10 ~]$ aws --version
aws-cli/2.22.26 Python/3.12.6 Linux/6.1.119-129.201.amzn2023.x86_64 exec-env/CloudShell exe/x86_64.amzn.2023
[cloudshell-user@ip-10-10-10-10 ~]$ aws iam list-users
{
  "Users": [
    {
      "Path": "/",
      "UserName": "Jazzy",
      "UserId": "AIDNOCZKIM:10-10-10-10",
      "Arn": "arn:aws:iam::1074-10-10-10:Jazzy",
      "CreateDate": "2024-12-30T19:13:20+00:00",
      "PasswordLastUsed": "2025-01-08T22:02:47+00:00"
    }
  ]
}
```

I wanted to venture back towards IAM to create a role for EC2 virtual servers.

What is a role: A role permits an entity to do stuff.

Select trusted entity Info

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.

Use case

- ☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.

Pretty similar experience to using Linux CMD. I created a demo file with 'test' as the string.

I printed my working directory > Clicked Actions > Download File > Put in

/home/cloudshell-user as the file path with /demo.txt at the end. I'll have to utilize CloudShell

more! I'm not a huge power user, but this is useful to have!

The screenshot displays the AWS CloudShell interface. On the left is a navigation sidebar with categories like 'Cost Categories', 'Budgets and Planning', and 'Savings and Commitments'. The 'Savings Plans' section is expanded. The main area shows a list of regions with checkboxes, including 'Israel (Tel Aviv)', 'Middle East (UAE)', and 'Asia Pacific'. A context menu is open over the 'Middle East (UAE)' region, showing 'File', 'Edit', and 'View' options. Below the region list is a 'CloudShell' terminal window for the 'us-east-2' region. The terminal shows the following commands and output:

```
[cloudshell-user@ip-10-0-10-10] $ echo "test" > demo.txt
[cloudshell-user@ip-10-0-10-10] $ cat demo.txt
test
[cloudshell-user@ip-10-0-10-10] $ pwd
/home/cloudshell-user
```

At the bottom right of the terminal window, it indicates 'Ln 1, Col 1' and '5 characters'.

I know that I'm going to venture more with EC2 instances, so I'm going to create a role for it in the future. I will come back to this!

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

DemoRoleForEC2

Maximum 64 characters. Use alphanumeric and '+, -, @, _' characters.

Description

Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+., @-/![]#\$%^&*()';"

Step 1: Select trusted entities

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"
12        ]
13      }
14    ]
15  }
16 }
```

Step 2: Add permissions

Permissions policy summary

Policy name	Type	Attached as
IAMReadOnlyAccess	AWS managed	Permissions policy

Roles (3) Info			
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.			
Search			
☐	Role name	☐ Trusted entities	Last activity
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linker)	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service)	-
☐	DemoRoleForEC2	AWS Service: ec2	-

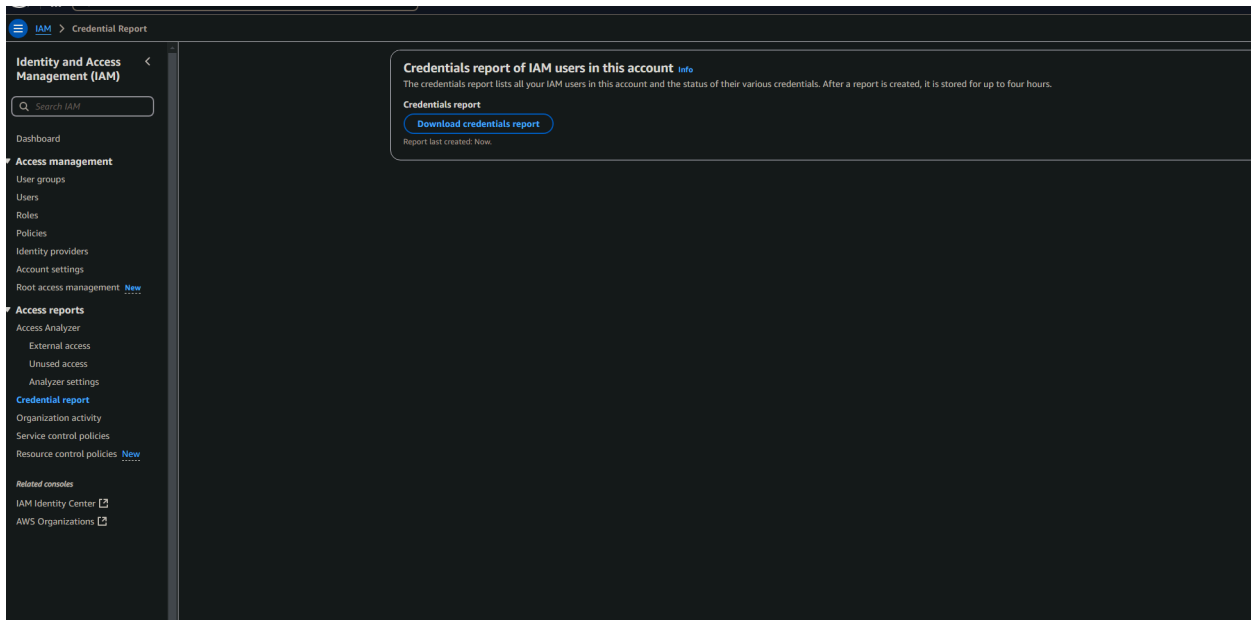
I'm a security-focused person, so I wanted to use IAM Credentials Report & IAM Access Advisor to monitor accounts and users.

Credentials Report is at the Account-level. It's a report list that lists all account's users and their status of their credentials.

Access Advisor is at a User-level and it shows the service permissions granted to a user and when those services were last accessed. This is basically Least-privilege

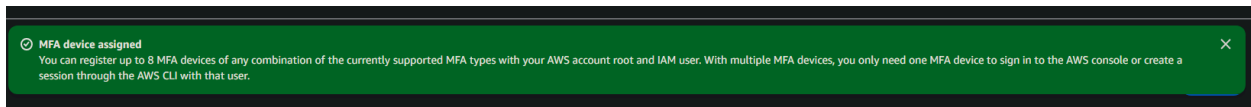
Both of these are great for auditing purposes and keeping maximum control over what a user works with. Misconfiguration within the cloud is a big thing this year.

Hands-On Creating a Credential Report



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
user	arn	user_crea	password	password	password	password	mfa_activ	access_ke	access_ke	access_ke	access_ke	access_ke	access_ke	access_ke	access_ke	access_ke	access_ke	cert_1_act	cert_1_las	cert_2_act	cert_2_last_rotated	
<root_	arn:aws:iam:2024-12-21	TRUE	2024-12-31	2024-12-21	not_supp	TRUE	FALSE	N/A	N/A	N/A	N/A	N/A	FALSE	N/A	N/A	N/A	N/A	FALSE	N/A	FALSE	N/A	
Jazzy	arn:aws:iam:2024-12-31	TRUE	2025-01-01	2024-12-31	N/A	FALSE	TRUE	2024-12-31	2024-12-31	us-east-1	iam		FALSE	N/A	N/A	N/A	N/A	FALSE	N/A	FALSE	N/A	

With this report, I have noticed that I do not have MFA enabled with my IAM account. This is the type of useful information I could use to enforce security policies



Access advisor is now recorded through Last Accessed.

The screenshot shows the 'Last Accessed' tab in the AWS IAM console. It displays a table of services accessed by the current user. The table has three columns: 'Service', 'Policies granting permissions', and 'Last accessed'. The 'Last accessed' column shows 'Today' for all entries. The services listed include AWS Organizations, AWS CloudShell, AWS IAM Identity Center, AWS Identity and Access Management, AWS IAM Access Analyzer, AWS Security Token Service, AWS Signin, AWS User Notifications, Amazon EC2, and AWS Support. There is a search bar and a filter dropdown at the top of the table. The page also includes a 'Permissions' tab and a 'Groups (1)' link in the top navigation bar.

Service	Policies granting permissions	Last accessed
AWS Organizations	AdministratorAccess	Today
AWS CloudShell	AdministratorAccess	Today
AWS IAM Identity Center (successor to AWS Single Sign-On)	AdministratorAccess	Today
AWS Identity and Access Management	IAMUserChangePassword and 1 more	Today
AWS IAM Access Analyzer	AdministratorAccess	Today
AWS Security Token Service	AdministratorAccess	Today
AWS Signin	AdministratorAccess	Today
AWS User Notifications	AdministratorAccess	Today
Amazon EC2	AdministratorAccess	Today
AWS Support	AdministratorAccess	Today

That’s pretty much a good introduction to IAM

I made sure to create another separate account. I don’t want to use the root account for everything.

I made some users and groups and assigned some permissions to said groups.

I created a very strong password policy.

I can check every account to see if they have MFA enabled

I created access keys through both the CLI

IAM Summary with terms

Users: Mapped to a physical user, has a password for AWS Console.

Groups: Contains users only.

Policies: Written in JSON documents that outline permissions for users or groups.

Roles: Entities used for EC2 instances and other AWS services. Think of it like a set group of assigned permissions.

AWS CLI - Manages your AWS services using the command line.

AWS SDK - Manages your AWS services using a programming language.

Access Keys: Gives access to AWS using the CLI or SDK.

Audits - IAM Credential Reports & IAM Access Advisor

Before I explore EC2, I want to make sure I'm not spending money. I want to use the free tier.

This will notify me if I ever go over my budget.

Choose budget type [info](#)

Budget setup

☒ **Use a template (simplified)**

Use the recommended configurations. You can change some configuration options after the budget is created.

☐ **Customize (advanced)**

Customize a budget to set parameters specific to your use case. You can customize the time period, the start month, and specific accounts.

Templates - new

Choose a template that best matches your use case.

☒ **Zero spend budget**

Create a budget that notifies you once your spending exceeds \$0.01 which is above the AWS Free Tier limits.

☐ **Monthly cost budget**

Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.

☐ **Daily Savings Plans coverage budget**

Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.

☐ **Daily reservation utilization budget**

Create a utilization budget for your reservations that notifies you when you fall below the defined target.

Zero spend budget - Template

Budget name

Provide a descriptive name for this budget.

My Zero-Spend Budget

Names must be between 1-100 characters.

Email recipients

Specify the email recipients you want to notify when the threshold has exceeded.

CamdenGowans77@gmail.com

Maximum number of email recipients is 10.

Scope

All AWS services are in scope in this budget.

☒ You will be notified via email when any spend above \$0.01 is incurred.

▼ Template settings

This template has default configurations that can be changed later. To change any of these settings, see [Custom](#). You can also download this template in [JSON](#).

[Cancel](#)

[Create budget](#)

Spending at max \$10

🔔 Your budget **My Zero-Spend Budget** has been created successfully. After creating a budget, it can take up to 24 hours to populate all of your spend data.

Submit feedback ✕

Choose budget type [Info](#)

Budget setup

☒ **Use a template (simplified)**

Use the recommended configurations. You can change some configuration options after the budget is created.

☐ **Customize (advanced)**

Customize a budget to set parameters specific to your use case. You can customize the time period, the start month, and specific accounts.

Templates - *new*

Choose a template that best matches your use case.

☐ **Zero spend budget**

Create a budget that notifies you once your spending exceeds \$0.01 which is above the AWS Free Tier limits.

☒ **Monthly cost budget**

Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.

☐ **Daily Savings Plans coverage budget**

Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.

☐ **Daily reservation utilization budget**

Create a utilization budget for your reservations that notifies you when you fall below the defined target.

Monthly cost budget - Template

Budget name

Provide a descriptive name for this budget.

My Monthly Cost Budget (\$10)

Names must be between 1-100 characters.

Enter your budgeted amount (\$)

Last month's cost: \$

10.00

Email recipients

Specify the email recipients you want to notify when the threshold has exceeded.

CamdenGowans77@gmail.com

Maximum number of email recipients is 10.

Scope

All AWS services are in scope in this budget

Hands-On Lab - Launching an EC2 Instance running Linux

I want to launch my first virtual server using the AWS Console. This is a web server that is launched using EC2 user data. This will give me an idea of how to start, stop, and terminate instances. The basics

AWS EC2 is flexible! It's cool how many different options I have at my disposal

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Quick Start

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu®

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

Debian
debian


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

A key pair is required to securely connect to your related instance. I'm going to create one.

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

EC2 Tutorial

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

 When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#) 

Cancel

Create key pair

I'll for sure dive into network related settings with EC2 instances. Just from looking at this, I'm noticing that my vpc IP is public. You can change it to private. You can create a security group, which is a set of firewall rules that control the traffic from your instance. The fun annoying Subnet stuff I can dive in with my networking knowledge later.

▼ Network settings [Info](#)

Edit

Network | [Info](#)

vpc-04-~~xxxxxx~~-a

Subnet | [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

With storage, AWS free tier can go up to 30 GB of EBS storage!

▼ **Configure storage** [Info](#)

Advanced

1x GiB Root volume 3000 IOPS (Not encrypted)

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Edit

This is a test, so I can just delete the volume after I terminate the EC2 instance

▼ **Storage (volumes)** [Info](#)

Simple

EBS Volumes [Hide details](#)

▼ Volume 1 (AMI Root)

Storage type [Info](#)

EBS

Device name - *required* [Info](#)

/dev/xvda

Snapshot [Info](#)

snap-0a4703089bc11d89d

Size (GiB) [Info](#)

Volume type [Info](#)

IOPS [Info](#)

Delete on termination [Info](#)

Encrypted [Info](#)

KMS key [Info](#)

Throughput [Info](#)

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

I want to launch a script right when the EC2 instance turns on. I have this:

```
#!/bin/bash
```

```
# install httpd (Linux 2 version)
```

```
yum update -y
```

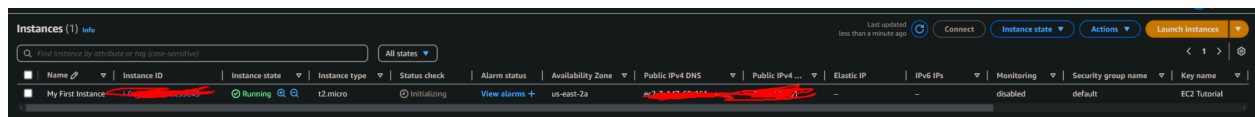
```
yum install -y httpd
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
echo "<h1>Hello World from $(hostname -f)</h1>" > /var/www/html/index.html
```

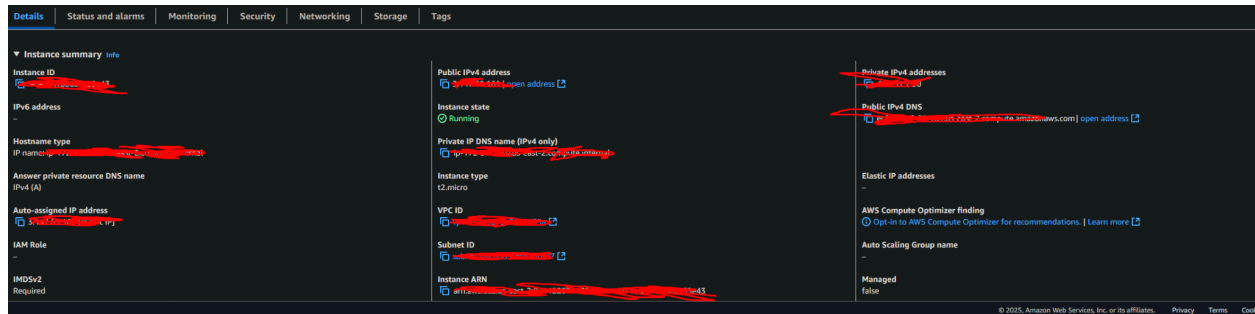
This script updates a system, and installs/starts the Apache web server. This will launch on boot and setup a basic homepage that displays a greeting with ‘Hello World’ with the server’s hostname. Super simple.



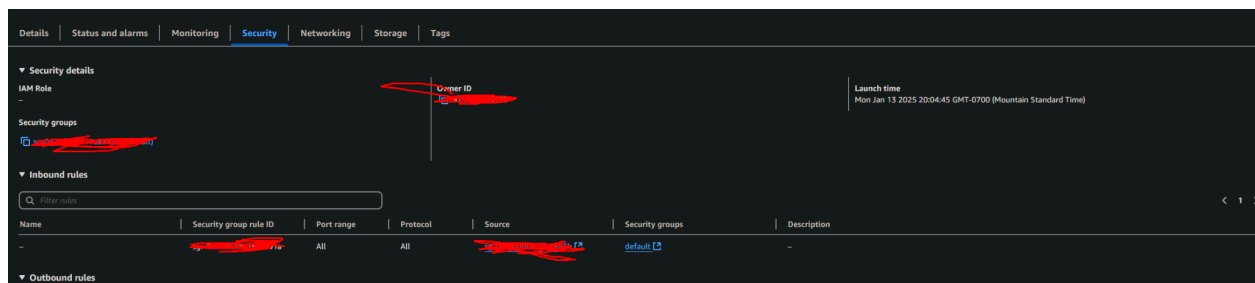
My instance is running and it took minimum time. This is the power of the cloud. I can spin up servers through AWS instead of being on-site in person to deal with this.

It converts the public IP to private with the power of NAT.

It also creates a DNS name with the IPv4 address!



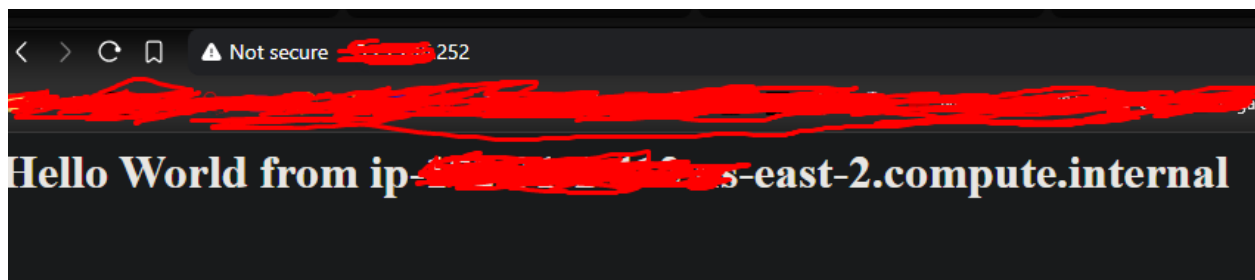
With the Security section, I will come back to this to configure security settings!



I have created my first Web server. When you click the Public IPv4 address, you need to do

http:\\ Public IP address

When you start and stop the IC2 instance, your public IP will change.



EC2 instance types:

General Purpose: Great for web servers or code repositories. Has a balanced approach between computing, memory, and networking

Compute-optimized Is Great for compute-intensive tasks that require high performance, such as batch processing workloads, media transcoding, high-performance web servers, High-performance computing, scientific modeling and machine learning, and dedicated gaming servers.

Memory Optimized: Fast performance for workloads that process large data sets in memory. High performance with relational/non-relational databases. Distributed web-scale cash stores, sued for Business intelligence and apps perform real-time processing for big unstructured data.

Storage Optimized: Great for storage-intensive tasks that require high, sequential read and write access to large data sets on local storage. Best used for high-frequency online transaction processing systems, relational and NoSQL databases, cache for in-memory databases, data warehousing applications, and distributed file systems.

Accelerated Computing - Use hardware accelerators, or co-processors, to perform functions, such as floating point number calculations, graphics processing, or data pattern matching.

HPC Optimized - built to offer the best price-performance for running HPC workloads at scale.

Ideal for applications that benefit from high-performance processors such as large, complex simulations and deep learning workloads.

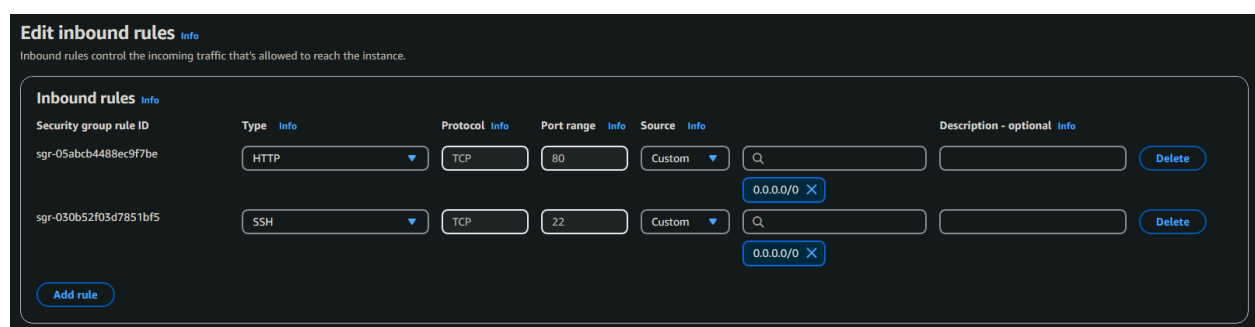
Let's work with a hands-on lab with security groups.

What is a security group?

Security groups control how traffic is allowed into or out of EC2 instances. They only contain allow rules, so it's a whitelist. They enforce group rules by IP or security group

Stateful by nature, if the inbound traffic is allowed on a port, the response is automatically allowed outbound. No explicitly deny. They have a couple of rules: Protocol (TCP, UDP, ICMP) Port range (80 for HTTP, 443 for HTTPS), source or Destination (IP range like 0.0.0.0/0 or even another security group).

For now, I have this set:



Let's play around with SSH. I download the security key of EC2 2 Tutorial.pem

I had to change some permissions, but changing to the location of the security key, which is on my desktop, I used `ssh -i '.\EC2 2 Tutorial.pem' ec2-user@ public IP`. That didn't work, so I had

to change permissions. Now, it's solely under my name and account. I am able to connect

[illegible]

I figured out that AWS has it's own version of SSH, it's called EC2 Instance Connect

Connect to instance [Info](#)

Connect to your instance **ec2-user@ip-10-0-10-10** (My Second Instance) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

 **ami-0a520a52** (My Second Instance)

Connection Type

☒ Connect using EC2 Instance Connect

Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ Connect using EC2 Instance Connect Endpoint

Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

☒ Public IPv4 address

 **10.0.10.10**

☐ IPv6 address

-

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

Super easy to go through this process.

```
aws [Alt+S]
[ec2-user@ip-172-31-1-10 ~]$ ping google.com
PING google.com (142.250.190.142) 56(84) bytes of data:
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=1 ttl=114 time=8.98 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=2 ttl=114 time=9.41 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=3 ttl=114 time=8.81 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=4 ttl=114 time=8.60 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=5 ttl=114 time=9.46 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=6 ttl=114 time=9.32 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=7 ttl=114 time=8.40 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=8 ttl=114 time=8.95 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=9 ttl=114 time=8.58 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=10 ttl=114 time=8.88 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=11 ttl=114 time=8.77 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=12 ttl=114 time=8.85 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=13 ttl=114 time=8.47 ms
64 bytes from ord37s36-in-f14.1e100.net (142.250.190.142): icmp_seq=14 ttl=114 time=8.83 ms
^C
--- google.com ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13022ms
rtt min/avg/max/mdev = 8.397/8.878/9.457/0.316 ms
[ec2-user@ip-172-31-1-10 ~]$
```

Within here, I want to view IAM roles. Let's try this:

```
[ec2-user@ip-172-31-1-10 ~]$ aws iam list-users

Unable to locate credentials. You can configure credentials by running "aws configure".
[ec2-user@ip-172-31-1-10 ~]$
```

I need to associate an IAM role with my EC2 instance:

Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID
i-008ce53f7892c8f6c (My Second Instance)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

Choose IAM role

Q |

No IAM Role
Choose this option to detach an IAM role

DemoRoleForEC2
arn:aws:iam::971422676179:instance-profile/DemoRoleForEC2

This worked;

```
[ec2-user@ip-172-31-2-112 ~]$ aws iam list-users
{
  "Users": [
    {
      "Path": "/",
      "UserName": "Jazzy",
      "UserId": "A1P5821F7892C8F6C56V",
      "Arn": "arn:aws:iam::971422676179:user/Jazzy",
      "CreateDate": "2024-12-30T19:13:20+00:00",
      "PasswordLastUsed": "2025-01-14T02:49:24+00:00"
    }
  ]
}
```

Why do you need to use IAM roles? To eliminate the need for hard-coded credentials. This role is attached to the EC2 instance and uses temporary security credentials. I don't have to manually rotate long-term credentials.

EC2 Spot Instances - They are the most cost-efficient instances in AWS. Useful for batch jobs, data analysis, image processing, and distributed workloads. These are not suitable for critical jobs or databases because you can get kicked out at any time.

Dedicated instances mean that you have your own instance on your own hardware.

Dedicated hosts mean you get access to the physical server itself and give you visibility to the lower-level hardware.

Good analogies:

- **On demand:** coming and staying in resort whenever we like, we pay the full price
- **Reserved:** like planning ahead and if we plan to stay for a long time, we may get a good discount.
- **Savings Plans:** pay a certain amount per hour for certain period and stay in any room type (e.g., King, Suite, Sea View, ...)
- **Spot instances:** the hotel allows people to bid for the empty rooms and the highest bidder keeps the rooms. You can get kicked out at any time
- **Dedicated Hosts:** We book an entire building of the resort
- **Capacity Reservations:** you book a room for a period with full price even you don't stay in it

Really nail down Spot Instances & Spot Fleet - On-Demand price

EC2 instance types nail down - Compute, storage, memory, general

Dedicated hosts, reserved instances

In all EC2 Instance Types and EC2 Purchasing options

Hands-On EC2 Placement Groups

A placement group is basically used to group multiple instances of EC2 instances.

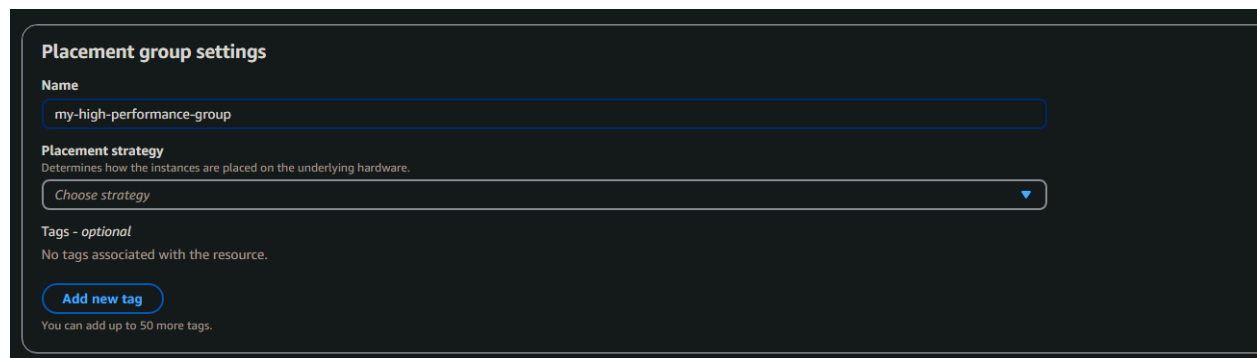
Types:

Cluster - All in the same AZ. Used for high network performance

Spread - Across different AZs with different hardware. Used for redundancy

Partition - Across different partitions, used with data stores and data replication.

Cluster



The screenshot shows the 'Placement group settings' panel in the AWS Management Console. It features a dark-themed interface with the following elements:

- Placement group settings**: The main title of the panel.
- Name**: A text input field containing the value 'my-high-performance-group'.
- Placement strategy**: A section header with a subtitle 'Determines how the instances are placed on the underlying hardware.' Below it is a dropdown menu currently showing 'Choose strategy' with a downward arrow.
- Tags - optional**: A section header with a subtitle 'No tags associated with the resource.' Below it is a button labeled 'Add new tag'.
- Footer note**: A small text line at the bottom stating 'You can add up to 50 more tags.'

Spread

Create placement group [Info](#)

Placement group settings

Name

Placement strategy
Determines how the instances are placed on the underlying hardware.

Spread ▼

Spread level
Determines how placement groups spread instances. You can only use host level spread placement groups on Outposts.

Rack (No restrictions) ▼

Tags - optional
No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Partition

Create placement group [Info](#)

Placement group settings

Name

Placement strategy
Determines how the instances are placed on the underlying hardware.

Partition ▼

Number of partitions
Choose the number of partitions to create in this placement group.

4 ▼

Tags - optional
No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

[Cancel](#) [Create group](#)

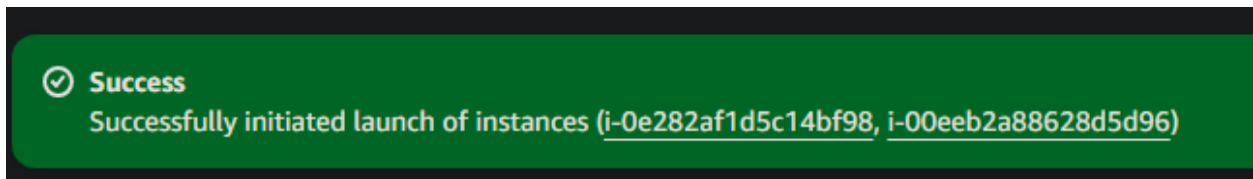
Now, I can put multiple EC2 instances within this placement group, which is a cluster. In that cluster, I could run high-performance use cases like HPC and big data operations.

Hands-on Creating Elastic Network Interfaces (ENI)

ENIs are virtual network interface cards that connect EC2 instances.

ENIs are bound to a specific AZ. You cannot attach an ENI to an EC2 instance in a different AZ

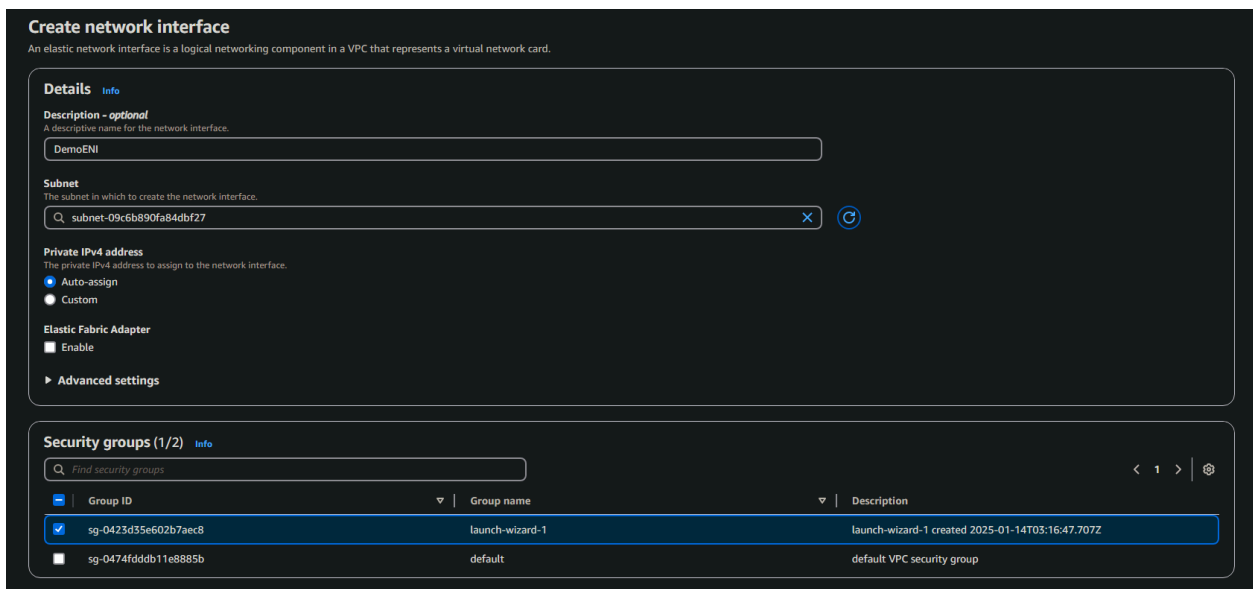
First, I need to create two instances



Currently using two different IPs. Let's change that.

ec2-18-191-147-140.us...	18.191.147.140
ec2-3-147-81-160.us-e...	3.147.81.160

Let's make sure they are in the same AZ



Interface Type	Description
Elastic network interface	–
Elastic network interface	–
Elastic network interface	<u>DemoENI</u>
Elastic network interface	–

Let's now attach the related ENI to the other instance

Things to study - EC2 Hibernate Root Volume type must be an Instance Store volume

Instance store volume vs. EBs volume

Attach network interface

Network interface

eni-0530534848a5a7572

VPC

vpc-04145a55e3688439a

Instance

i-0e282af1d5c14bf98

▼ ENA Express - optional

ⓘ

ENA Express

The selected instance type (t2.micro) does not support ENA Express.

ENA Express

Enable ENA Express to increase the maximum bandwidth used per stream and minimize tail latency of network traffic between EC2 instances.

☐ Enable

ENA Express UDP

Use ENA Express for UDP traffic.

☐ Enable

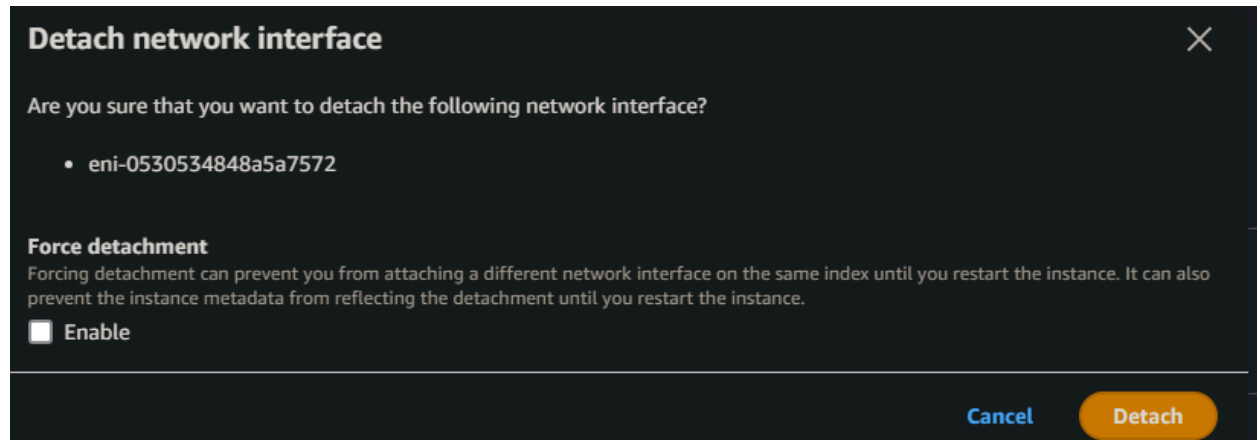
Cancel

Attach

This is now attached.

▼ Network Interfaces (2) Info									
Filter network interfaces									
Interface ID	Device index	Card index	Description	Public IPv4 address	Private IPv4 address	Private IPv4 DNS	IPv6 addresses	Primary IPv6 address	
 eni-0530534848a5a7572	1	0	DemoENI	–	172.31.7.120	ip-172-31-7-120.us-...	–	–	
 eni-06eae7fb45f6e4cc9	0	0	–	3.147.81.160	172.31.8.22	ip-172-31-8-22.us-e...	–	–	

Let's say I want to do some network failover tests. I can detach this ENI



Let's attach this to a new instance with the same ENI

Attach network interface

Network interface

eni-0530534848a5a7572

VPC

vpc-04145a55e3688439a

Instance

i-00eeb2a88628d5d96

▼ ENA Express - optional

i

ENA Express

The selected instance type (t2.micro) does not support ENA Express.

ENA Express

Enable ENA Express to increase the maximum bandwidth used per stream and minimize tail latency of network traffic between EC2 instances.

☐ Enable

ENA Express UDP

Use ENA Express for UDP traffic.

☐ Enable

Cancel

Attach

EC2 instances can be saved in-memory. You need to make sure the EBS root volume is encrypted. This makes it really fast and seamless to go back to that related instances once you put it to sleep.

Important note: You do not pay for the compute, but you do pay for the storage

Encrypting the Root Volume

▼ **Storage (volumes)** [Info](#) Simple

EBS Volumes Hide details

▼ Volume 1 (AMI Root) (Custom)

Storage type Info EBS	Device name - <i>required</i> Info /dev/xvda	Snapshot Info snap-01adb1609d1f61341
Size (GiB) Info 8	Volume type Info gp3 ▼	IOPS Info 3000
Delete on termination Info Yes ▼	Encrypted Info Encrypted ▼	KMS key Info (default) aws/ebs Key ID: alias/aws/ebs ▼
Throughput Info 125		

① Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage ×

[Add new volume](#)

Instance enabled to allow my EC2 instance to hibernate

Learn more'."/>

Stop - Hibernate behavior | [Info](#)

Enable ▼

To enable hibernation, space is allocated on the root volume to store the instance memory (RAM). Make sure that the root volume is large enough to store the RAM contents and accommodate your expected usage, e.g. OS, applications. To use hibernation, the root volume must be an encrypted EBS volume. [Learn more](#)

Testing EC2 Hibernation

Now, let's hibernate the instance

Now, let's hibernate the instance

Private IPv4 addresses
Public IPv4 DNS
Elastic IP addresses
AWS Compute Optimizer finding
Auto Scaling Group name
Managed
false

Connect Instance state Actions

Stop instance
Start instance
Reboot instance
Hibernate instance
Terminate (delete) instance

EC2 Hibernate... i-0fa6d6da68f4a4dff Stopped t2.micro

Let's start the instance and see if the uptime is still going

```
05:20:59 up 4 min, 1 user, load average: 0.15, 0.23, 0.10
```

EBS & EBS Snapshots Hands-on

What is EBS - It's an Elastic Block Store, a network drive you can attach to your instances while they run. It allows your instance to save data even after terminating it. They can only be mounted to one instance at a time and they are bound to a specific availability zone. It's basically a network USB drive

I created an EC2 instance

volumes (1/1) Info

Saved filter sets

Choose filter set

Q Search

⌛

Actions

Create volume

< 1 >

⚙

☒	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	Alarm status
☒	-	vol-08b1eea8682cc9d06	gp3	8 GiB	3000	125	snap-0a47030...	2025/01/15 21:32 GMT-7	us-east-2a	✔ In-use	No alarms

Now, let's add a volume to this. Let's make it a general-purpose SSD gp2 with 2 GiB. I made sure it's in the same AZ as my instance with us-east-2a, and I enabled encryption

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type | [Info](#)
General Purpose SSD (gp2) ▼

Size (GiB) | [Info](#)
2
Min: 1 GiB, Max: 16384 GiB.

IOPS | [Info](#)
100 / 3000
Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.

Throughput (MiB/s) | [Info](#)
Not applicable

Availability Zone | [Info](#)
us-east-2a ▼

Snapshot ID - optional | [Info](#)
Don't create volume from a snapshot ▼ 

Encryption | [Info](#)
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☒ Encrypt this volume

KMS key | [Info](#)
(default) aws/ebs ▼ 

KMS key description

Volumes (1/2) [Info](#)

Saved filter sets
Choose filter set ▼

Search

	Name ▼	Volume ID ▼	Type ▼	Size ▼
☐	—	vol-08b1eea8682cc9d06	gp3	8 GiB
☒	—	vol-0e5e4f127f5350396	gp2	2 GiB

Now, I need to attach the volume

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

This volume is encrypted and it can only be attached to an instance that supports EBS encryption. [Learn more](#)

Volume ID
 vol-0e5e4f127f5350396

Availability Zone
us-east-2a

Instance [Info](#)

i-0f0c22ab75e45cbfd
(My EC2 Instance) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)

/dev/sdf

Recommended device names for Linux: /dev/xvda for root volume. /dev/sd[f-p] for data volumes.

Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

▼ Root device details

Root device name

 /dev/xvda

Root device type

EBS

EBS encryption

disabled

▼ Block devices

Q

Filter block devices

	Volume ID	Device name	Volume size (GiB)	Attachment status	Attachment time	Encrypted
	vol-08b1eea8682cc9d06	/dev/xvda	8	 Attached	2025/01/15 21:32 GMT-7	No
	vol-0e5e4f127f5350396	/dev/sdf	2	 Attached	2025/01/15 21:38 GMT-7	Yes

Pretty straightforward process with all of this! I can easily add more space if I want to within Volumes.

Further on, EBS Snapshots. These are backupS of EBS volumes at a point in time.

EBS Snapshot Archive - You can move a snapshot to an archive tier that is 75% cheaper. It takes within 24 to 72 hours for restoring the archive.

Recycle Bin - EBS Snapshots - You can setup rules to retain delated snapshots that you can recover. Great for accidental deletions. You can specify the retention from 1 day to 1 year.

Fast Snapshot Restore (FSR) - Force full initialization of the snapshot to have no latency on the first use. Costs a lot of money though.

Let's create a snapshot of my 8 GiB volume

Create snapshot Info

Create a point-in-time snapshot of an EBS volume and use it as a baseline for new volumes or for data backup. You can create snapshots from an individual volume, or you can create multi-volume snapshots from all of the volumes attached to an instance.

Source

Resource type | Info

☒ Volume
Create a snapshot from a specific volume.

☐ Instance
Create multi-volume snapshots from an instance.

Volume ID
The volume from which to create the snapshot.

vol-08b1eea8682cc9d06
us-east-2a

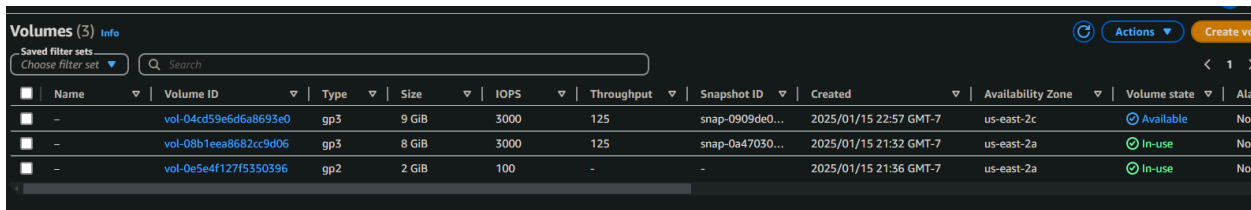
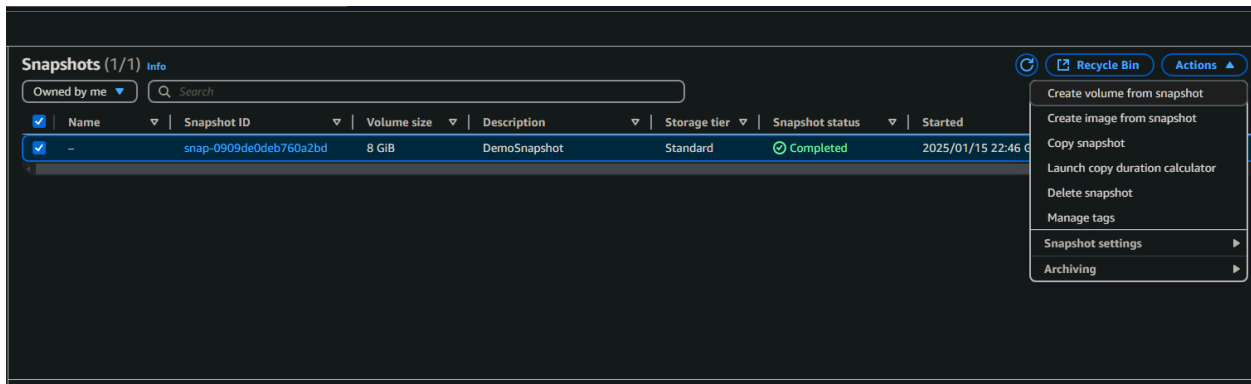
Snapshot details

Description
Add a description for your snapshot.

DemoSnapshot

Name	Snapshot ID	Volume size	Description	Storage tier	Snapshot status	Started	Progress
-	snap-0909de0deb760a2bd	8 GiB	DemoSnapshot	Standard	Completed	2025/01/15 22:46 GMT-7	100%

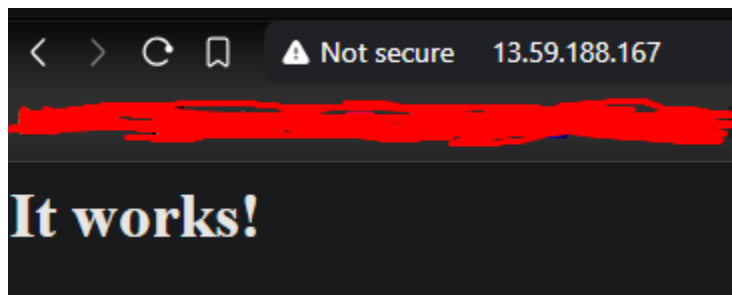
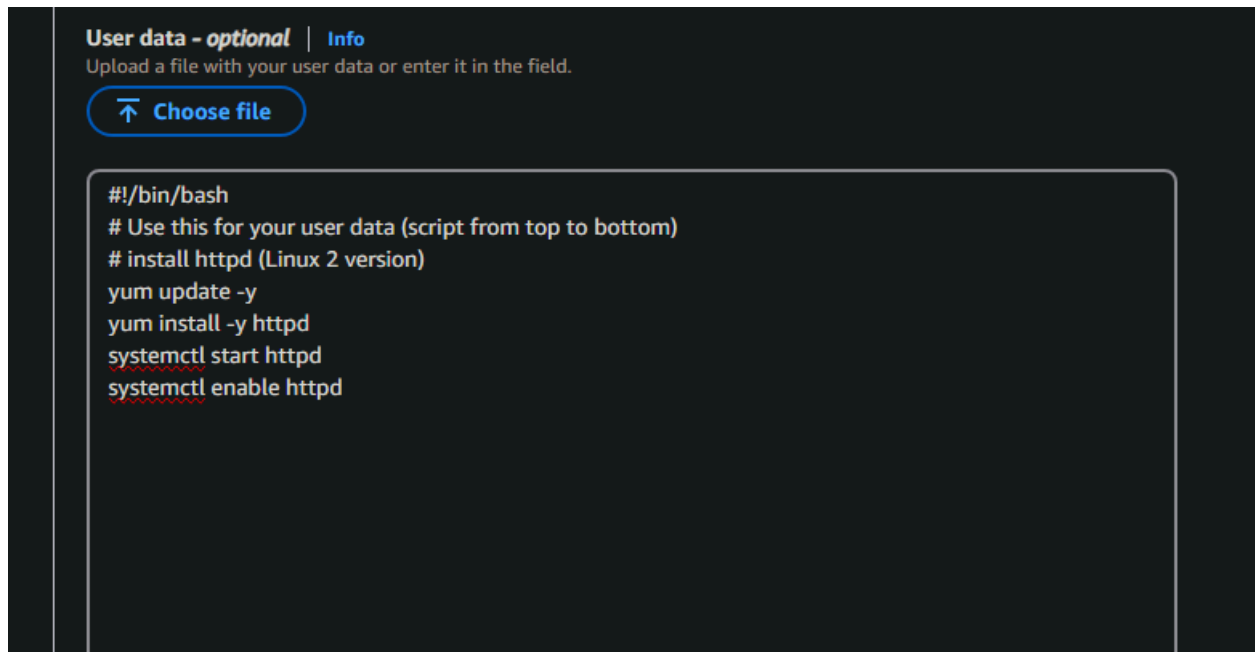
Let's create a volume from the related snapshot. This would be great for Disaster Recovery. I put the snapshot in a different AZ - us-east-2c



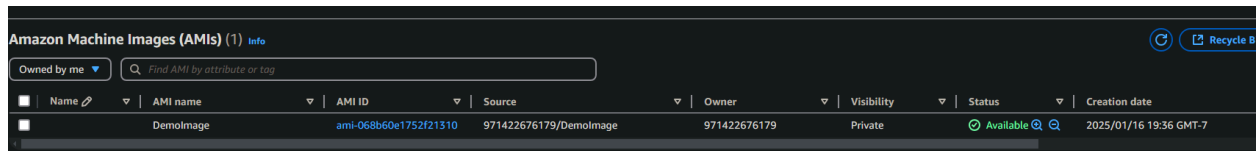
Hands-on Amazon Machine Images (AMIs)

What is an AMI? It's basically a custom built EC2 instances that have configurations, applications, services, etc.

Let's launch this and have it launch an Apache server

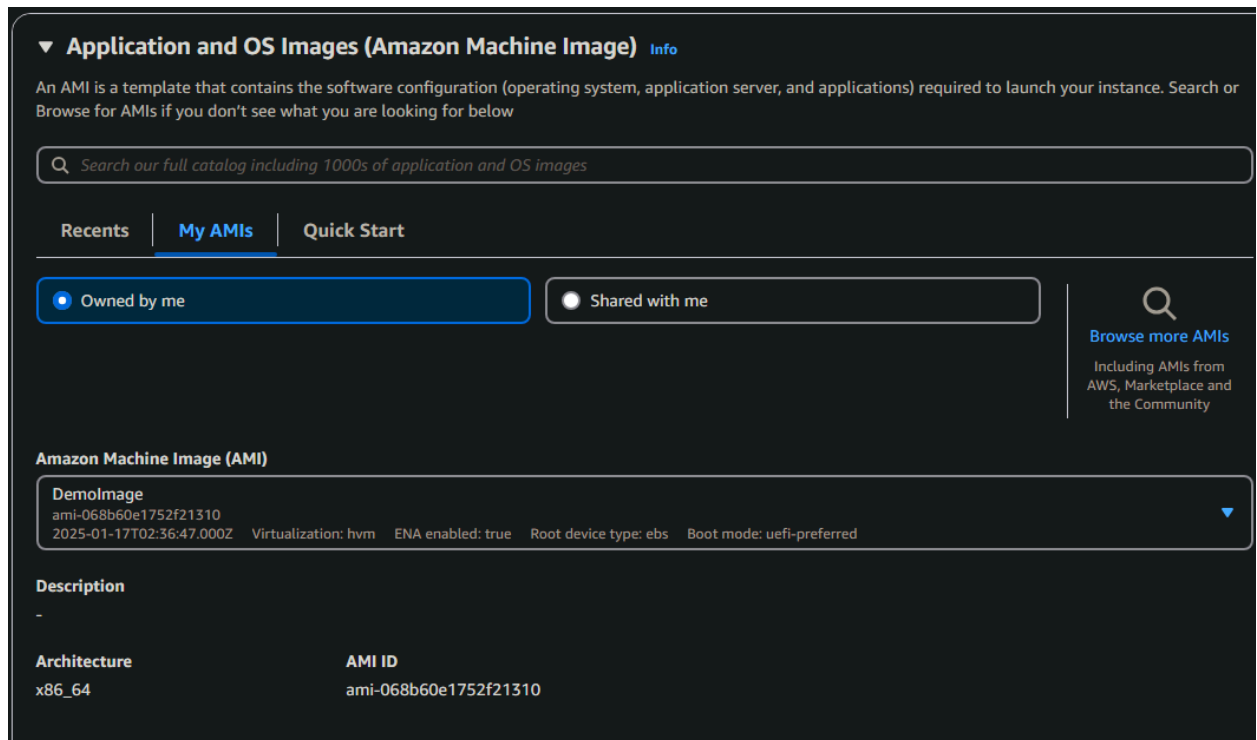


Now, let's create an image



	Name	AMI name	AMI ID	Source	Owner	Visibility	Status	Creation date
	DemolImage		ami-068b60e1752f21310	971422676179/DemolImage	971422676179	Private	Available	2025/01/16 19:36 GMT-7

Cool, now I have that. Let's create a new EC2 instance based on that image



Application and OS Images (Amazon Machine Image) Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents | **My AMIs** | Quick Start

☒ Owned by me ☐ Shared with me

Amazon Machine Image (AMI)

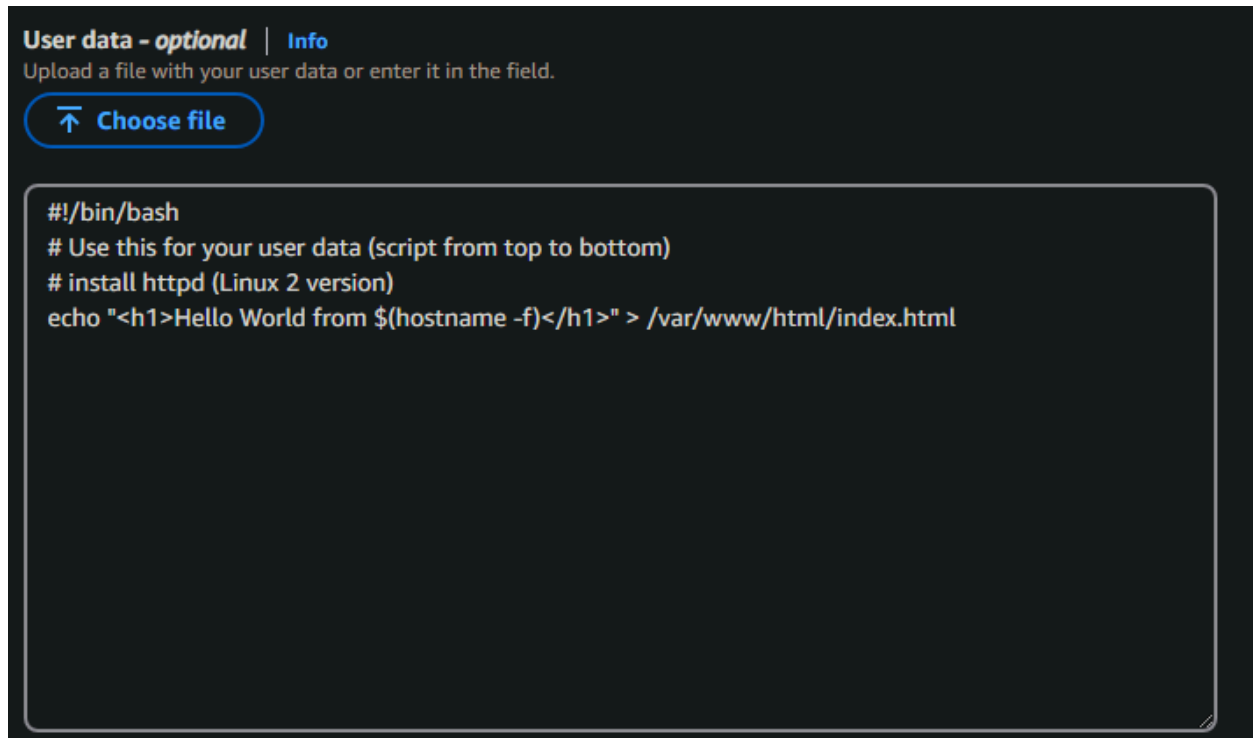
DemolImage
ami-068b60e1752f21310
2025-01-17T02:36:47.000Z Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred

Description

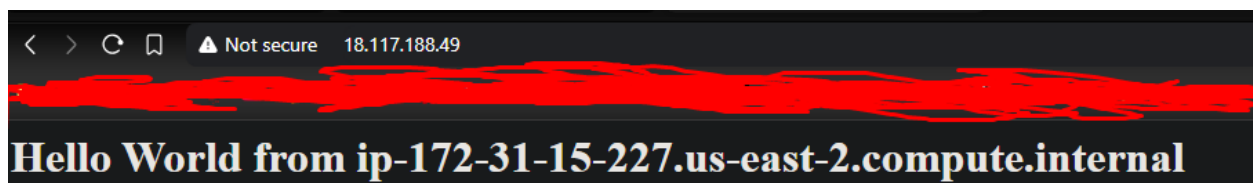
-

Architecture x86_64 **AMI ID** ami-068b60e1752f21310

Now, I already have the Apache server installed within the image. Let's just print: "Hello World from". I don't need the yum update and install.



Done



Very powerful tool. From a cybersecurity perspective, I could install security tools like KACE, Nessus Agent, Splunk Agent, EDR solutions, etc., thus making it really easy to replicate it for many machines. I just have to spend time configuring one host.

Hands-On Encrypting a non-encrypted EBS volume

I want to play around with encryption with EBS

Scenario: Let's say I create an unencrypted EBS volume to store data backed up from a different AZ. I want to encrypt that encrypted EBS volume. Let's try this.

I'll create an EBS snapshot of a volume that is not encrypted.

Create volume info

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type info
General Purpose SSD (gp3) 

Size (GiB) info
1
Min: 1 GiB, Max: 16384 GiB.

IOPS info
3000
Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s) info
125
Min: 125 MiB, Max: 1000 MiB, Baseline: 125 MiB/s.

Availability Zone info
us-east-2a 

Snapshot ID - optional info
Don't create volume from a snapshot 

Encryption info
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☒ Encrypt this volume

Tags - optional info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add tag

You can add 50 more tags.

Snapshot summary info

 Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.



I created a snapshot of said volume that is also not encrypted.

Snapshots (1/1) Info

Owned by me

Search

☒	Name	Snapshot ID	Volume size	Description	Storage tier	Snapshot status	Started	Progress	Encryption
☒	-	snap-0d4d30274e8e6cedb	8 GiB	Created by CreateImagefj-...	Standard	Completed	2025/01/16 19:38 GMT-7	100%	Not encrypted

Snapshot ID: snap-0d4d30274e8e6cedb

Details

Snapshot settings

Storage tier

Tags

Snapshot ID

☐ snap-0d4d30274e8e6cedb

Started

☐ Thu Jan 16 2025 19:38:18 GMT-0700 (Mountain Standard Time)

▼ Source volume

Volume ID

☐ vol-03d567c62959f18bf

▼ Encryption

Encryption

Not encrypted

Progress

🟢 100%

Product codes

-

Snapshot status

🟢 Completed

Fast snapshot restore

-

Volume size

☐ 8 GiB

KMS key ID

-

KMS key alias

-

Now, I copy that said snapshot and enable encryption with a KMS key

Copy snapshot Info

Copy a snapshot from one AWS Region to another, or within the same Region.

Source snapshot

The original snapshot that is to be copied.

Snapshot ID

 snap-0d4d30274e8e6cedb

Region

us-east-2

Snapshot copy details

Description

A description for the snapshot copy.

[Copied snap-0d4d30274e8e6cedb from us-east-2] Created by CreateImage(i-0ca8374b87919cab2) for ami-068b60e1752f21310

255 characters maximum.

Destination Region

The Region in which to create the snapshot copy.

us-east-2

Time-based copy - *new* | [Info](#)

Specify a completion duration for the snapshot copy operation. Additional costs apply. [Learn more](#) 

☐ Enable time-based copy

Encryption | [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources.

☒ Encrypt this snapshot

KMS key | [Info](#)

(default) aws/ebs



KMS key description

 Default key that protects my EBS volumes when no other key is defined

KMS key owner account

 971422676179 (This account)

KMS key ID

 1150fd99-3c4b-45af-802d-84338e4157ed

KMS key ARN

 arn:aws:kms:us-east-2:971422676179:key/1150fd99-3c4b-45af-802d-84338e4157ed

Now, let's create a volume using that newly created encrypted snapshot

snap-0988e9b6c5e3c2b39

Snapshot ID snap-0988e9b6c5e3c2b39	Progress 0%
Started Thu Jan 16 2025 20:32:41 GMT-0700 (Mountain Standard Time)	Product codes -
Source volume Volume ID vol-ffffff	Volume size 8 GiB
Encryption Encrypted	KMS key ID 1150fd99-3c4b-45af-802d-84338e4157ed

Snapshot settings

Storage tier

Tags

Snapshot Lock - new
Lock mode
Not locked

Share permissions
Snapshot share permissions
Private
The snapshot is shared only with AWS accounts that you specified.

You can only share snapshots that are encrypted with a customer managed key.

Here is what I did: I created an EBS volume that is not encrypted. I created a snapshot of the EBS volume that is also not encrypted. I copied that snapshot and enabled encryption with the KMS key. Finally, I created a volume off that secure copy snapshot.

If I want to do this faster, I can also create an encrypted volume using an encrypted snapshot

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Snapshot ID
 `snap-0d4d30274e8e6cedb`

Volume type | [Info](#)
General Purpose SSD (gp3) ▼

Size (GiB) | [Info](#)

Min: 1 GiB, Max: 16384 GiB.

IOPS | [Info](#)

Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s) | [Info](#)

Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Availability Zone | [Info](#)
us-east-2a ▼

Fast snapshot restore | [Info](#)
Not enabled for selected snapshot

Encryption
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☒ **Encrypt this volume**

KMS key | [Info](#)
 ▼ 

Hands-On Create Elastic File System (EFS)

An EFS is a managed network file system that can be mounted on many EC2 instances. They do not have to be in the same AZ. It is highly available and it automatically scales per-pay-use.

Use cases: content management, web serving, data sharing, WordPress

Let's create a file system. With that lifecycle management, this is where I would have a set policy.

File system settings

General

Name - optional
Name your file system.

File system type
Choose to either store data across multiple Availability Zones or within a single Availability Zone. [Learn more](#)

☒ **Regional**
Offers the highest levels of availability and durability by storing file system data across multiple Availability Zones within an AWS Region.

☐ **One Zone**
Provides continuous availability to data within a single Availability Zone within an AWS Region.

Automatic backups
Automatically backup your file system data with AWS Backup using recommended settings. Additional pricing applies. [Learn more](#)

☒ **Enable automatic backups**

Lifecycle management
Automatically save money as access patterns change by moving files into the Infrequent Access (IA) or Archive storage class. [Learn more](#)

Transition into Infrequent Access (IA)
Transition files to IA based on the time since they were last accessed in Standard storage.

Transition into Archive
Transition files to Archive based on the time since they were last accessed in Standard storage.

Transition into Standard
Transition files back to Standard storage based on when they are first accessed in IA or Archive storage.

Encryption
Choose to enable encryption of your file system's data at rest. Uses the AWS KMS service key (aws/elasticfilesystem) by default. [Learn more](#)

☒ **Enable encryption of data at rest**

[Customize encryption settings](#)

I need to create a specific security group for my EFS through EC2 - Security Group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Security groups

▼

sg-014c1e7a1bb36ef30 ✕
efs-demo

Remove

▼

sg-014c1e7a1bb36ef30 ✕
efs-demo

Remove

▼

sg-014c1e7a1bb36ef30 ✕
efs-demo

Remove

This is now created

Amazon EFS > File systems > fs-0982fed4605416b8b

fs-0982fed4605416b8b Delete Attach

General Edit

Amazon resource name (ARN)
arn:aws:elasticfilesystem:us-east-2:971422676179:file-system/fs-0982fed4605416b8b

Performance mode
General Purpose

Throughput mode
Elastic

Lifecycle management
Transition into Infrequent Access (IA): 30 day(s) since last access
Transition into Archive: 90 day(s) since last access
Transition into Standard: None

Availability zone
Regional

Automatic backups
Enabled

Encrypted
cadd13a2-bd27-4665-a910-22aa8b754548 (aws/elasticfilesystem)

File system state
Available

DNS name
fs-0982fed4605416b8b.efs.us-east-2.amazonaws.com

Replication overwrite protection
Enabled

Metered size | Monitoring | Tags | File system policy | Access points | Network | Replication

Metered size

Total size
6.00 KiB

Size in Standard
6.00 KiB (100%)

Size in IA
0 Bytes (0%)

Size in Archive
0 Bytes (0%)



6.00 KiB

Size in Standard Size in IA Size in Archive

Now, let's mount an EC2 instance to this EFS. This is in east a

▼ **Configure storage** Info Advanced

1x GiB ▼ Root volume 3000 IOPS (Not encrypted)

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage ×

Add new volume

Click refresh to view backup information ↻
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems Edit

This is already really flexible, because it automatically attaches security groups to enable access to the file system, and it automatically mounts that shared file system.

Let's create another Instance in east b

File systems

Hide details

● EFS

● FSx

▼ Shared file system 1

Remove

File system | Info

Mount point | Info

fs-0982fed4605416b8b
Availability: Regional

/mnt/efs/fs1

Add shared file system

Create new shared file system

4 remaining (Up to 5 file systems maximum).

☒ Automatically create and attach security groups

To enable access to the file system, the required security groups will be automatically created and attached to this instance and the selected file system. To manually manage the security groups, clear the checkbox. [Learn more.](#)

☒ Automatically mount shared file system by attaching required user data script

Automatically mount your file system by updating your user data to install efs-utils. If you would like to manually mount your file system, clear the checkbox.

Now I have two instances located in different availability zones

Instances (2) Info							
Find Instance by attribute or tag (case-sensitive)				All states ▼			
☐	Name ↗ ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status	Availability Zone ▼
☐	Instance B	i-036dbe8068be6d93a	Running 🔍 🔍	t2.micro	2/2 checks passed	View alarms +	us-east-2b
☐	Instance A	i-0a912670f135a99ac	Running 🔍 🔍	t2.micro	2/2 checks passed	View alarms +	us-east-2a

So now, if I go back to the EFS I created earlier, I can see that it's replicated against different AZs with multiple security groups

Network							
Availability zone (AZ-ID)	Mount target ID	Subnet ID	Mount target state	IP address	Network interface ID	Security groups	
us-east-2a (use2-az1)	fsmt-00778ab36671f6324	subnet-09c6b890fab4dbf27	Available	172.31.13.202	eni-09fd205e52f5820c	sg-030425c65d7cc1f33 (efs-sg-1), sg-014c1e7a1bb36ef30 (efs-demo), sg-089c42688ac9980ec (efs-sg-2)	
us-east-2b (use2-az2)	fsmt-01cde1d4835a1c58a	subnet-0efc4901be4e39445	Available	172.31.27.3	eni-010f576c5bfcdb34b	sg-030425c65d7cc1f33 (efs-sg-1), sg-014c1e7a1bb36ef30 (efs-demo), sg-089c42688ac9980ec (efs-sg-2)	
us-east-2c (use2-az3)	fsmt-0c51e1dac0bc679ca	subnet-0df7bb237c7cb3dee	Available	172.31.38.70	eni-096cd950fce1e3360	sg-030425c65d7cc1f33 (efs-sg-1), sg-014c1e7a1bb36ef30 (efs-demo), sg-089c42688ac9980ec (efs-sg-2)	

This inbound group within EC2 is connected to the security group as well

Inbound rules (1)							
Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	
-	sg-0888789dcb840328b	-	NFS	TCP	2049	sg-0db89aebb08aed01...	

Let's now connect to both instances through EC2

To verify everything working, let's create a file in Instance A

```
~/mnt
[ec2-user@ip-172-31-21-142 ~]$ sudo su
[root@ip-172-31-21-142 ec2-user]# echo "hello world! Cloud security is fun, yet moving" > /mnt/efs/fs1/hello.txt
[root@ip-172-31-21-142 ec2-user]# ^C
[root@ip-172-31-21-142 ec2-user]# cat /mnt/efs/fs1/hello.txt
hello world! Cloud security is fun, yet moving
[root@ip-172-31-21-142 ec2-user]#
```

Now, let's go over to Instance B and list out the created file

[illegible]

Scalability and High Availability

Scalability means that an application/system can handle greater loads by adapting.

Vertical Scalability - Increasing the size of the instance. AWS Example, Increasing the instance size. Useful for databases and non-distributed systems. RDS.

Horizontal Scalability - Increase the amount of instances/systems for your applications. Distributed systems.

High Availability - Being able to run your application/system in at least 2 data centers - availability zones. Done through Auto Scaling groups and load balancers.

Load Balancer AWS types:

Classic Load Balancer (v1 - old generation)

Application Load Balancer (v2 - new generation) HTTP, HTTPS, WebSocket

Network Load Balancer (v2 - new generation) TCP, TLS, UDP

Gateway Load Balancer - 2020 GWLB (Network layer) IP protocol

Hands-On Creating an Application Load Balancer (ALB)

Let's generate some traffic by creating two Instances

Instances (2/2) [Info](#)

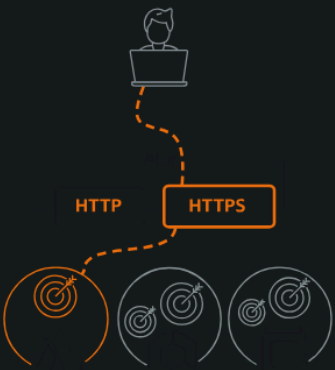
Find Instance by attribute or tag (case-sensitive) All states ▼

☒	Name 	Instance ID	Instance state 	Instance type 	Status check
☒	My First Instance	i-0c1f19c4a83f9de86	 Running  	t2.micro	 Initializing
☒	My Second Ins...	i-00ccffb86e1c45e9c	 Running  	t2.micro	 Initializing

Let's create an application load balancer

Load balancer types

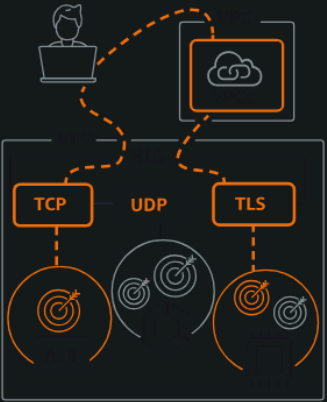
Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

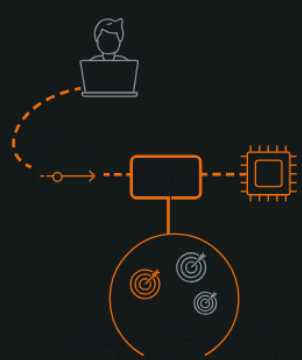
Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create

Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

► Classic Load Balancer - *previous generation*

Close

Now, I will deploy it across 3 AZs

Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#). For a new VPC, [create a VPC](#).

-

vpc-04145a55e3688439a
IPv4 VPC CIDR: 172.31.0.0/16

Mappings Info
Select at least two Availability Zones and one subnet per zone. The load balancer routes traffic to targets in these Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

Availability Zones

☒ **us-east-2a (use2-az1)**
Subnet

subnet-09c6b890fa84dbf27
IPv4 subnet CIDR: 172.31.0.0/20

IPv4 address
Assigned by AWS

☒ **us-east-2b (use2-az2)**
Subnet

subnet-0efc4901be4e39445
IPv4 subnet CIDR: 172.31.16.0/20

IPv4 address
Assigned by AWS

☒ **us-east-2c (use2-az3)**
Subnet

subnet-0df7bb237c7cb3dee
IPv4 subnet CIDR: 172.31.32.0/20

IPv4 address
Assigned by AWS

Let's now create a new security group to allow HTTP traffic

Inbound rules Info

Type Info

Protocol Info

Port range Info

Source Info

Description - optional Info

HTTP

TCP

80

Anywhere...

Q 0.0.0.0/0

Delete

0.0.0.0/0 X

Add rule

Outbound rules Info

Type Info

Protocol Info

Port range Info

Destination Info

Description - optional Info

All traffic

All

All

Custom

Q 0.0.0.0/0 X

Delete

Add rule

Security group (sg-02011b73ea06d3676 | demo-sg-load-balancer) was created successfully
Details

sg-02011b73ea06d3676 - demo-sg-load-balancer

Actions

Details

Security group name
demo-sg-load-balancer

Security group ID
sg-02011b73ea06d3676

Description
Allow HTTP into ALB

VPC ID
vpc-04145a55e3688439a

Owner
971422676179

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-03f80b7d16b7e6c49	IPv4	HTTP	TCP	80	0.0.0.0/0	-

Now I can add this to the Create Application Load Balancer page

Security groups

Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

demo-sg-load-balancer
sg-02011b73ea06d3676 VPC: vpc-04145a55e3688439a

Now, let's create a target group

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

Instances

• Supports load balancing to instances within a specific VPC.

• Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

IP addresses

• Supports load balancing to VPC and on-premises resources.

• Facilitates routing to multiple IP addresses and network interfaces on the same instance.

• Offers flexibility with microservice based architectures, simplifying inter-application communication.

• Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

Lambda function

• Facilitates routing to a single Lambda function.

• Accessible to Application Load Balancers only.

Application Load Balancer

• Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.

• Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

demo-tg-alb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation

HTTP

80

1-65535

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Instance ID

Name

State

Security groups

Zone

Private IPv4 address

Subnet ID

Launch time

i-0c1f19c4a83f9de86

My First Instance

Running

launch-wizard-1

us-east-2a

172.31.6.101

subnet-09c6a890fa84dbf27

January 22, 2025, 22:41 (UTC-07:00)

i-00ccfb86e1c45e9c

My Second Instance

Running

launch-wizard-1

us-east-2a

172.31.4.101

subnet-09c6a890fa84dbf27

January 22, 2025, 22:41 (UTC-07:00)

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with comma)

Include as pending below

demo-tg-alb

Actions

Details

arn:aws:elasticloadbalancing:us-east-2:971422676179:targetgroup/demo-tg-alb/d9bf4616a5057e9a

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-04145a55e3688439a

IP address type

IPv4

Load balancer

None associated

2

Total targets

0

Healthy

0

Anomalous

0

Unhealthy

2

Unused

0

Initial

0

Draining

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 >

Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overrid...	Launch ...	Anomaly detection...
i-00ccffb86e1c45e9c	My Second Inst...	80	us-east-2a (us...	Unused	Target group is not co...	-	-	January 2...	Normal
i-0c1f19c4a83f9de86	My First Instance	80	us-east-2a (us...	Unused	Target group is not co...	-	-	January 2...	Normal

Listeners and routing

Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Remove

Protocol

Port

HTTP

80

1-65535

Default action

Info

Forward to

demo-tg-alb

Target type: Instance, IPv4

HTTP

Create target group

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Add listener

Nice, this is now created

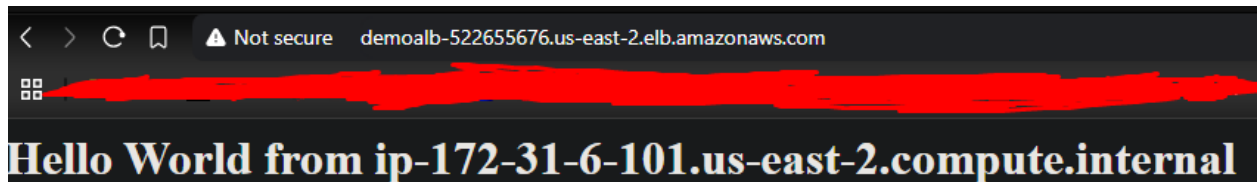
Load balancers (1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

Name	DNS name	State	VPC ID	Availability Zones	Type	Date created
DemoALB	DemoALB-522655676.us-e...	Active	vpc-04145a55e3688439a	3 Availability Zones	application	January 22, 2025, 22:51 (UTC-07:00)

Instead of accessing the usual server via the public IP, I can now use the DNS name to go to the website. That's one of many cool features with ALBs.



That's not all, I can now refresh this page and look what happens



It switches between instances per refresh! I can also use this as a health check to see how well the load balancer is working with the instances

demo-tg-alb

Actions

Details

arn:aws:elasticloadbalancing:us-east-2:971422676179:targetgroup/demo-tg-alb/d9bf4616a5057e9a

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-04145a55e3688439a
IP address type IPv4	Load balancer DemoALB		

2 Total targets	2 Healthy 0 Anomalous	0 Unhealthy	0 Unused	0 Initial	0 Draining
--------------------	-----------------------------	----------------	-------------	--------------	---------------

► **Distribution of targets by Availability Zone (AZ)**
Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets | Monitoring | Health checks | Attributes | Tags

Registered targets (2) Info

Anomaly mitigation: Not applicable | Deregister | Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

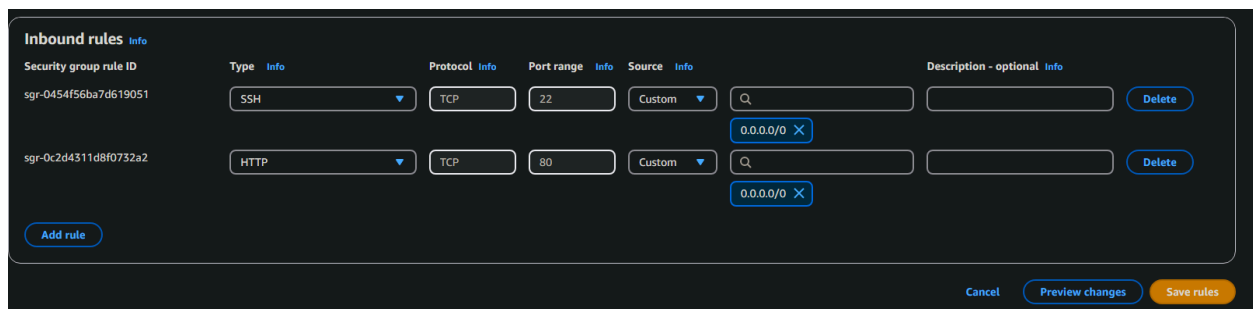
Filter targets

Instance ID	Name	Port	Zone	Health status	Health status details	Administrative o...	Override details	Launch...	Anomaly
i-00ccf8b8e1c45e9c	My Second Inst...	80	us-east-2a (us...	Healthy	-	No override	No override is curren...	January 2...	Normal
i-0c1f19c4a83f9de86	My First Instance	80	us-east-2a (us...	Healthy	-	No override	No override is curren...	January 2...	Normal

Hands-On Securing ALB

We created an ALB last lab. It is open to all traffic, which isn't secure at all. Let's lock this down.

With the launch-wizard-1 security group, which I created when I launched the My Instance 1 EC2, I can adjust the inbound rules.



I want to configure traffic to be only accessed from the load balancer. Let's delete the HTTP inbound rule. What's really cool is that you can determine traffic not only from a network, you can configure it to match the relating security group!

Inbound rules

Info

Security group rule ID

sgr-0454f56ba7d619051

Type

Info

SSH

Protocol

Info

TCP

Port range

Info

22

Source

Info

Custom

Description - optional

Info

Q

0.0.0.0/0

X

Delete

Add rule

Q

Delete

changes

Save rules

CIDR blocks

0.0.0.0/0

0.0.0.0/8

0.0.0.0/16

0.0.0.0/24

0.0.0.0/32

:/0

:/16

:/32

:/48

:/64

Security Groups

instance-sg-2 | sg-0db89aebb08aed014

efs-sg-1 | sg-030425c65d7cc1f33

launch-wizard-1 | sg-0dec881e19f62fc70

efs-sg-2 | sg-089c42dd8a0980e0c

instance-sg-1 | sg-0c55eb710237636a9

demo-sg-load-balancer | sg-02011b73ea06d3676

default | sg-0474fdddb11e8885b

demo-sg-load-balancer | sg-02011b73ea06d3676

To verify this now, I can try to launch the website through the public IP.

This will continue to load



This site can't be reached

18.225.254.238 took too long to respond.

Try:

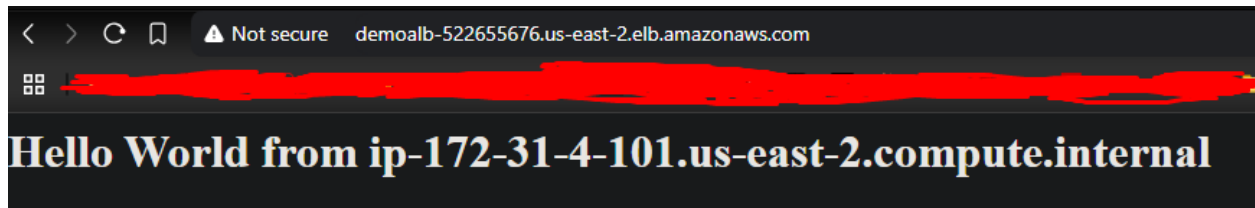
- Checking the connection
- Checking the proxy and the firewall

ERR_CONNECTION_TIMED_OUT

Reload

Details

Now, let's try the DNS name. This works as intended



Now, let's say I want to connect to this website through the DNS name with a separate page with /error. Let's play around with this.

First, I need to adjust the listener rules within the Listeners and Rules - Load balancer. Let's return a fixed response

A screenshot of the AWS Load Balancing console's "Response" configuration page. The page has two main sections: "Response code" and "Response body - optional".
- The "Response code" section has a label "The type of message you want to send." and a text input field containing "404". Below the field is a small hint: "2xx, 4xx, 5xx".
- The "Content type - optional" section has a label "The format of your message." and a dropdown menu currently set to "text/plain".
- The "Response body - optional" section has a label "Enter your response message." and a large text area containing "Not found, custom error!". Below the text area is a hint: "1024 character maximum".
At the bottom right of the form are three buttons: "Cancel", "Previous", and "Next".

Now, I can try reaching the website with that /error at the end

Not found, custom error!

Hands-On Network Load Balancer (v2)

This is at Layer 4 and it forwards TCP & UDP traffic to instances. These can handle millions of requests per second and has ultra-low latency. NLB has one static IP per AZ, and supports assigning elastic IPs

Let's create a NLB

Mapped to three different AZs

Mappings
Select one or more Availability Zones and corresponding subnets. Enabling multiple Availability Zones increases the fault tolerance of your applications. The load balancer routes traffic to targets in the same Availability Zone as the load balancer or the VPC are not available for selection.

Availability Zones

☒ **us-east-2a (use2-az1)**

Subnet

subnet-09c6b890fa84dbf27
IPv4 subnet CIDR: 172.31.0.0/20

IPv4 address
The front-end IPv4 address of the load balancer in the selected Availability Zone.

☒ Assigned by AWS ☐ Use an Elastic IP address

☒ **us-east-2b (use2-az2)**

Subnet

subnet-0efc4901be4e39445
IPv4 subnet CIDR: 172.31.16.0/20

IPv4 address
The front-end IPv4 address of the load balancer in the selected Availability Zone.

☒ Assigned by AWS ☐ Use an Elastic IP address

☒ **us-east-2c (use2-az3)**

Subnet

subnet-0df7bb237c7cb3dee
IPv4 subnet CIDR: 172.31.32.0/20

IPv4 address
The front-end IPv4 address of the load balancer in the selected Availability Zone.

☒ Assigned by AWS ☐ Use an Elastic IP address

It is recommended to create a security group with NLBs

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

Name cannot be edited after creation.

Description Info

VPC Info

vpc-04145a55e3688439a

Inbound rules Info

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
HTTP	TCP	80	Anywh...	Q 0.0.0.0/0	Delete
0.0.0.0/0					
Add rule					

Outbound rules Info

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info	
All traffic	All	All	Custom	Q	Delete
0.0.0.0/0					
Add rule					

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups - recommended

Security groups support on Network Load Balancers can only be enabled at creation by including at least one security group. You can change security groups after creation. The security groups for your load balancer must allow it to communicate with registered targets on both the listener port and the health check port. For PrivateLink Network Load Balancers, security group rules are enforced on PrivateLink traffic; however, you can turn off inbound rule evaluation after creation within the load balancer's Security tab or using the API.

Select up to 5 security groups

demo-sg-nlb

sg-0f97bf51b67406d4 VPC: vpc-04145a55e3688439a

X

Let's create a target group

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

☒ Instances

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

☐ IP addresses

- Supports load balancing to VPC and on-premises resources.
- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

☐ Lambda function

- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

☐ Application Load Balancer

- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

demo-tg-nlb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation

TCP

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

-
vpc-04145a55e3688439a
IPv4 VPC CIDR: 172.31.0.0/16

Successfully created the target group: demo-tg-nlb.

demo-tg-nlb

Actions

Details

arn:aws:elasticloadbalancing:us-east-2:971422676179:targetgroup/demo-tg-nlb/38909039d81d1650

Target type

Instance

Protocol : Port

TCP: 80

VPC

vpc-04145a55e3688439a

IP address type

IPv4

Load balancer

None associated

Total targets

2

Healthy

0

Unhealthy

0

Unused

2

Initial

0

Draining

0

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Filter targets

Deregister

Register targets

<

1

>

	Instance ID	Name	Port	Zone	Health status	Health status det...	Administrative ...	Override details	Launch time
	i-09e19259ac5b2...	My Second Instance	80	us-east-2a (use2-...	Unused	Target group is no...	-	-	January 23, 2025,...
	i-04e95a23500efc...	My First Instance	80	us-east-2a (use2-...	Unused	Target group is no...	-	-	January 23, 2025,...

Listeners and routing

Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener TCP:80

Remove

Protocol

TCP

Port

80

1-65535

Default action

Info

Forward to

demo-tg-nlb

TCP

Target type: Instance, IPv4

Create target group

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Add listener

Good to go

DemoNLB

▼ Details

Load balancer type

Network

Scheme

Internet-facing

Status

✔ Active

Hosted zone

ZLMOA37VPKANP

Load balancer ARN

 `arn:aws:elasticloadbalancing:us-east-2:971422676179:loadbalancer/net/DemoNLB/35e614b726d753e4`

Hands-On Auto Scaling Groups

Auto Scaling Groups are used to design resilient, high-performing, and cost-optimized architectures. It's used to handle varying workloads while maintaining availability and performance. It adjusts EC2 instances based on conditions

First, I need to create a launch template

It's going to be an Amazon Linux instance

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ Application and OS Images (Amazon Machine Image) - required [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI

ami-018875e7376831abe (64-bit (x86), uefi-preferred) / ami-031bb1452a43b486d (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

▼ Instance type [Info](#) [Get advice](#) [Advanced](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

Free tier eligible ▼

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

EC2 2 Tutorial

▼

[Create new key pair](#)

This launch instance is going to start up a Linux web server. What's nice, is that it creates the website in different instances.

Now, I need to create an Auto Scaling Group based off that launch template.

Choose launch template Info

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name
Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

Launch template Info

ⓘ For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

[Create a launch template](#)

Version

[Create a launch template version](#)

Description Template	Launch template MyDemoTeplate lt-0921438ceb379c0fa	Instance type t2.micro
AMI ID ami-018875e7376831abe	Security groups -	Request Spot Instances No
Key pair name EC2 2 Tutorial	Security group IDs sg-0dec881e19f62fc70	

Additional details

Storage (volumes) -	Date created Tue Feb 04 2025 21:22:04 GMT-0700 (Mountain Standard Time)
-------------------------------	---

Cancel Next

Now, it is going to launch against 3 different availability zones

Choose instance launch options Info

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements Info

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template	Version	Description
MyDemoTemplate ↗ lt-0921438ceb379c0fa	Default	Template

Instance type
t2.micro

Network Info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-04145a55e3688439a
172.31.0.0/16 Default [↕](#)

[Create a VPC ↗](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets [↕](#)

us-east-2a | subnet-09c6b890fa84dbf27 [✕](#)
172.31.0.0/20 Default

us-east-2b | subnet-0efc4901be4e39445 [✕](#)
172.31.16.0/20 Default

us-east-2c | subnet-0df7bb237c7cb3dee [✕](#)
172.31.32.0/20 Default

[Create a subnet ↗](#)

Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

Let's configure it with a load balancer

Integrate with other services - *optional* [Info](#)

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ No load balancer

Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ Attach to an existing load balancer

Choose from your existing load balancers.

☐ Attach to a new load balancer

Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.

☒ Choose from your load balancer target groups

This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

demo-tg-alb | HTTP

Load balancer: Not associated with any load balancer

ⓘ One of your target groups is not yet associated with any load balancer. In order for routing and scaling to occur, you will need to attach the target group to a load balancer. This can be done later in the [Load Balancing console](#). [↗](#)

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

ⓘ Always enabled

Additional health check types - *optional* | [Info](#)

☒ Turn on Elastic Load Balancing health checks Recommended

Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

ⓘ EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing. To avoid unexpected terminations, first verify the settings of these health checks in the [Load Balancer console](#). [↗](#)

☐ Turn on VPC Lattice health checks

VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks

EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period | [Info](#)

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300

seconds

Done

Auto Scaling groups (1) Info								
Search your Auto Scaling groups								
☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones
☐	DemoASG	MyDemoTemplate Version Default	0	⏸ Updating capacity...	1	1	1	us-east-2a, us-east-2b, us-east-2c

Activity history (1)				
Filter activity history				
Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-0d63482f0e4089008	At 2025-02-05T04:31:07Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 1. At 2025-02-05T04:31:21Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 1.	2025 February 04, 09:31:23 PM -0700	2025 February 04, 09:31:29 PM -0700

```

      #_
    ~\  ##### Amazon Linux 2023
   ~~ \  #####\
  ~~  \#####|
  ~~   \###|
  ~~    \|/
  ~~~~~ V~' '->
        ~~~
         ~~-.-
          /-/-
           /m/' -
[ec2-user@ip-172-31-43-13 ~]$
```

Now, let's scale it up

In my DemoASG, I can edit the Group size

Instance management

Instance refresh

Activity

Monitoring

Group size

×

Specify the size of the Auto Scaling group by changing the desired capacity. You can also specify minimum and maximum scaling limits.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ▾

Desired capacity

Specify your group size.

2 ▴ ▾

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

2

Equal or greater than desired capacity

Cancel

Update

Activity history (2)

Filter activity history

< 1 > ⚙

Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-0c9e991411fbdc6fa	At 2025-02-05T04:35:46Z a user request update of AutoScalingGroup constraints to min: 1, max: 2, desired: 2 changing the desired capacity from 1 to 2. At 2025-02-05T04:35:50Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 1 to 2.	2025 February 04, 09:35:51 PM -07:00	2025 February 04, 09:35:58 PM -07:00
Successful	Launching a new EC2 instance: i-0d63482f0e4089008	At 2025-02-05T04:31:07Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 1. At 2025-02-05T04:31:21Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 1.	2025 February 04, 09:31:23 PM -07:00	2025 February 04, 09:31:29 PM -07:00

Instances (2)

Filter instances

< 1 > ⚙

Actions ▾

Instance ID	Lifecycle	Instance type	Weighted capacity	Launch template...	Availability Zone	Health status	Protected from
i-0c9e991411fbdc6fa	InService	t2.micro	-	MyDemoTemplate Ver	us-east-2a	Healthy	
i-0d63482f0e4089008	InService	t2.micro	-	MyDemoTemplate Ver	us-east-2c	Healthy	

Hands-on Auto Scaling Groups - Scaling Policies

Let's create a dynamic scaling policy

I'm going to have a control in place. If the CPU utilization goes over 40%, I want to add more EC2 instances.

Target Tracking Policy☐

Policy type
Target tracking scaling

Enabled or disabled
Enabled

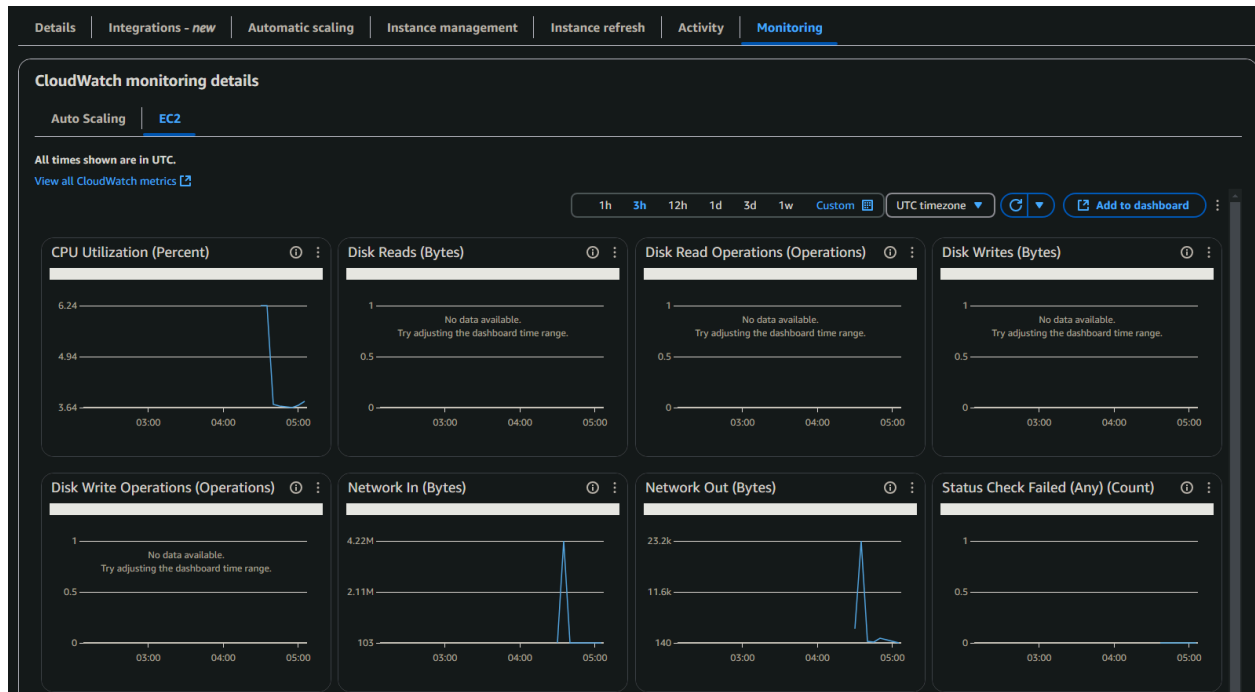
Execute policy when
As required to maintain Average CPU utilization at 40

Take the action
Add or remove capacity units as required

Instances need
300 seconds to warm up before including in metric

Scale in
Enabled

Now, I want to see this in action

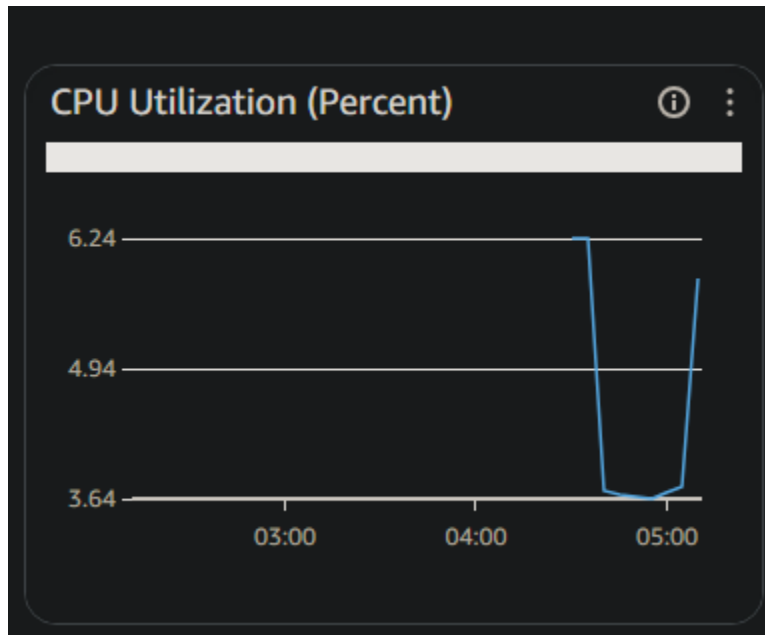


Let's stress the EC2 instance

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

ec2-user@ip-172-31-15-246:~$ sudo amazon-linux-extras install epel -y
sudo: amazon-linux-extras: command not found
ec2-user@ip-172-31-15-246:~$ sudo yum install stress -y
Last metadata expiration check: 0:34:16 ago on Wed Feb  5 04:36:25 2023.
No match for argument: stress
Error: Unable to find a match: stress
ec2-user@ip-172-31-15-246:~$ sudo yum install stress -y
Last metadata expiration check: 0:34:21 ago on Wed Feb  5 04:36:25 2023.
Dependencies resolved.
Package                               Architecture      Version           Repository        Size
Installing:
stress                               x86_64            1.0.7-2.amzn2023.0.1  amazonlinux      34 k
Transaction Summary
Install 1 Package
Total download size: 34 k
Installed size: 48 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm                                940 kb/s | 34 kB | 00:00
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      |
  Installing    | stress-1.0.7-2.amzn2023.0.1.x86_64
  Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64
  Verifying     | stress-1.0.7-2.amzn2023.0.1.x86_64
  Installing    | stress-1.0.7-2.amzn2023.0.1.x86_64
Complete!
ec2-user@ip-172-31-15-246:~$
```

Increasing



1 instance before

Instances (1)								
Filter instances								
	Instance ID	Lifecycle	Instance type	Weighted capacity	Launch template...	Availability Zone	Health status	Protected from
	i-0c9e991411fbd66fa	InService	t2.micro	-	MyDemoTeplate	us-east-2a	Healthy	

2 instances now after stressing the CPU

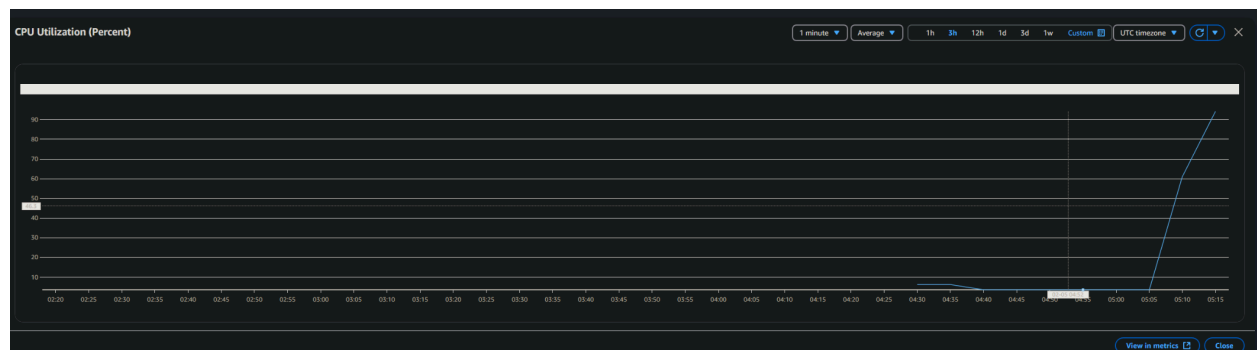
Instances (2)								
Filter instances								
	Instance ID	Lifecycle	Instance type	Weighted capacity	Launch template...	Availability Zone	Health status	Protected from
	i-04cf26590e9b2b5a2	InService	t2.micro	-	MyDemoTeplate	us-east-2c	Healthy	
	i-0c9e991411fbd66fa	InService	t2.micro	-	MyDemoTeplate	us-east-2a	Healthy	

In CloudWatch, I can see the alarm. It gives me the activity here

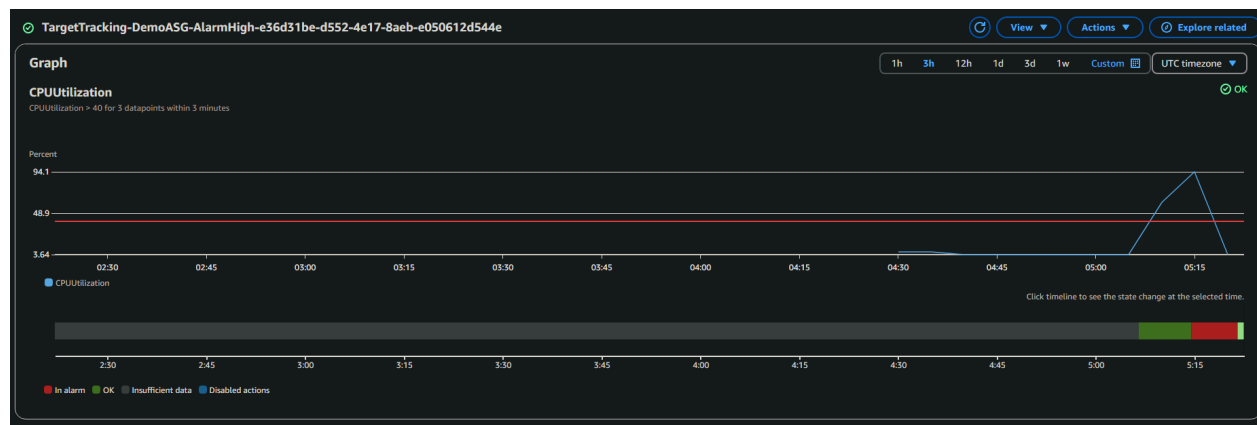


Now, let's reboot the instance that stressed the CPU

High up right now



Now that the CPU utilization is down. It worked as intended



Hands-on RDS Read Replicas

First, let's create a My SQL Database

Create database [info](#)

Choose a database creation method

☒ **Standard create**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [info](#)

☐ Aurora (MySQL Compatible)


☒ MySQL


☐ MariaDB


☐ Microsoft SQL Server


☐ Aurora (PostgreSQL Compatible)


☐ PostgreSQL


☐ Oracle


☐ IBM Db2


Let's create a security group

DB subnet group [info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access [info](#)

☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ **Choose existing**
Choose existing VPC security groups

☒ **Create new**
Create new VPC security group

New VPC security group name

demo-database-mysql

Availability Zone [info](#)

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ **Create an RDS Proxy** [info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

It’s cool that it gives you the estimated monthly costs at the bottom.

Estimated monthly costs

DB instance	12.41 USD
Storage	2.30 USD
Total	14.71 USD

This billing estimate is based on on-demand usage as described in [Amazon RDS Pricing](#). Estimate does not include costs for backup storage, I/Os (if applicable), or data transfer.

Estimate your monthly costs for the DB Instance using the [AWS Simple Monthly Calculator](#).

Creating database database-1

Your database might take a few minutes to launch. You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

View credential details

database-1

Summary

DB identifier

database-1

Status

Creating

Role

Instance

Engine

MySQL Community

Recommendations

CPU

Class

db.t3.micro

Current activity

Region & AZ

us-east-2c

Let’s connect to this using SQLElectron

database-1

Summary

DB identifier

database-1

Status

Modifying

Role

Instance

Engine

MySQL Community

Recommendations

CPU

27.32%

Class

db.t3.micro

Current activity

0 Connections

Region & AZ

us-east-2c

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations - new

Tags

Recommendations

Connectivity & security

port

point copied

database-1.c300644psic3.us-east-2.rds.amazonaws.com

Networking

Availability Zone

us-east-2c

VPC

Security

VPC security groups

demo-database-mysql (sg-0208cc0f2a48e205c)

Active

Sqlectron

FileQueryEditViewFindHelp

SQLELECTRON

servers

Search

No results

Settings

Server Information

Connection Test

Successfully connected

Name

RDSDEMO

Database Type

MySQL

SSL

Server Address

database-1.c3006k4qsic3.us-east-2

3306

Domain

Unix socket path

User

admin

Password

••••••••

Initial Database/Keyspace

mydb

Initial Schema

Schema

URI

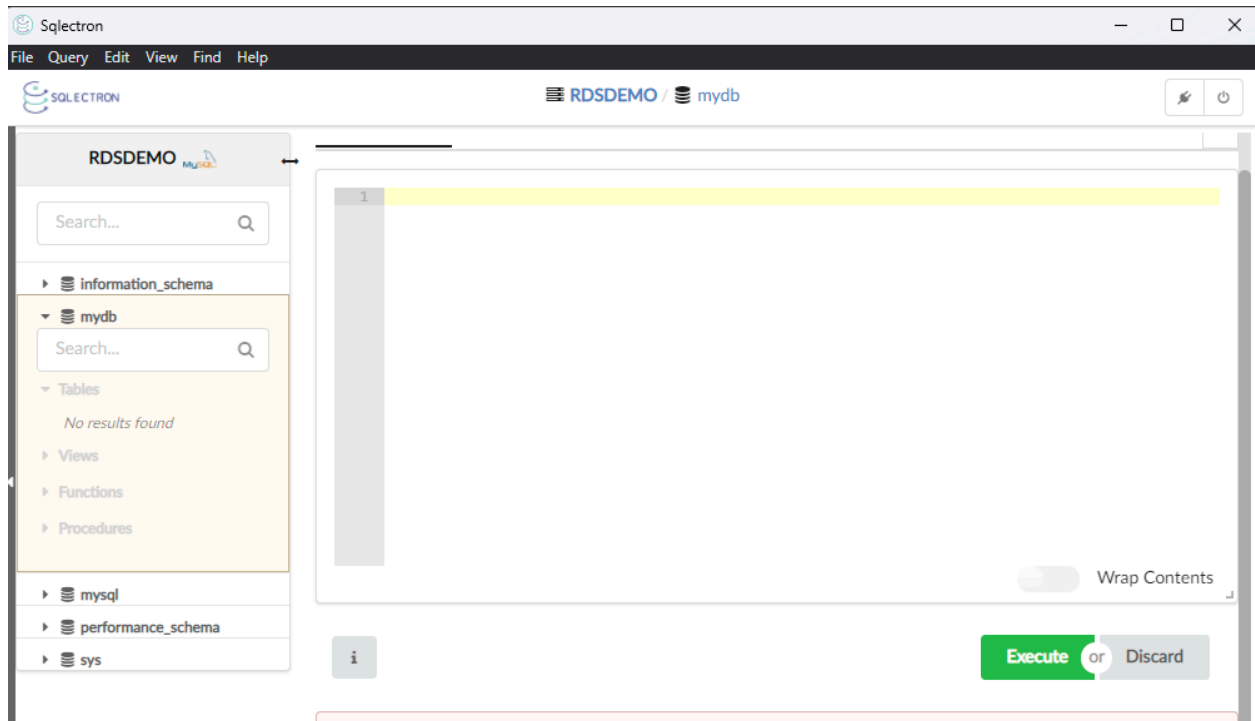
mysql://admin:*****@database-1.c3006k4qsic3.us-east-2.rds.amazonaws.com/mydb

Make the password visible in order to change the database credentials through the URI format.

SSH Tunnel

Filter

Sweet, this is now created



Aurora supports Postgres and MySQL for databases. Your drivers will work as if Aurora were a Postgres or MySQL database. It's cloud-optimized and it has 5x performance improvement over MySQL on RDS, and over 3x the performance of Postgres on RDS. It automatically grows in increments of 10GB, up to 128 TB. It also stores 6 copies of your data across 3 AZs. It's self-healing with peer-to-peer replication. Storage is striped across 100s of volumes

Writer Endpoint - Connects to the primary DB Instance (handles writes & reads). Used for applications that need to write data. Automatically redirects to the new primary if a failover occurs.

Reader Endpoint - Load-balances connections across all Aurora Replicas (read-only). Used for scaling read-heavy workloads (analytics and reporting). Automatically adjusts if replicas are added/removed.

Custom Endpoint - User-defined endpoint that routes to a subset of DB instances. Used to isolate workloads - direct reporting queries to low-priority replicas.

Endpoint - Default endpoint that connects to the current primary instance.

Instance Endpoint - Directly connects to a specific DB instance. Used for debugging or maintenance.

Aurora Serverless - Automated database instantiation and auto-scaling based on actual usage.

Good for infrequent or intermittent or unpredictable workloads

If you stop an RDS database, you will still pay for storage. If you plan on stopping it for a long time, you should snapshot and restore instead

Hands-on ElastiCache

Let's create an ElastiCache

Cache settings Info

Amazon ElastiCache announces Valkey starting as low as \$6/month

Introducing Valkey, an open source project that is a drop in replacement of Redis OSS, now available on ElastiCache, with up to 33% lower price than other engines. [Learn more](#)

Try ElastiCache (Valkey)

Cache engine

Select the engine that will run in your cache

Engine selected
Redis OSS

Choose Valkey engine

Benefits	Redis OSS	Valkey (Learn more)
Latest ElastiCache features & innovation	⊖	✓
Price (Serverless)	+49% higher than Valkey	Save 33% compared to Redis OSS
Price (Node based)	+25% higher than Valkey	Save 20% compared to Redis OSS
Higher throughput, memory efficiency	⊖	✓
Continued open source innovation	⊖	✓

Configuration Info

Choose one of the following options to create a Redis cache.

Deployment option

☐ Serverless - *new*
Use to quickly create a cache that automatically scales to meet application traffic demands, with no servers to manage.

☒ Design your own cache
Use to create a cache by selecting node type, size, and count.

Creation method

☐ Easy create
Use recommended best practice configurations. You can also modify options after you create the cluster.

☒ Cluster cache
Set all of the configuration options for your new cluster.

☐ Restore from backup
Use an existing backup or .rdb file to restore a cluster.

Cluster mode

Scale your cluster dynamically with no downtime.

☐ Enabled
Cluster mode enables replication across multiple shards for enhanced scalability and availability.

☒ Disabled
The Redis OSS cluster will have a single shard (node group) with one primary node and up to 5 read replica.

Done

Advanced settings [Info](#)

Security
Use the following section to configure network security and data security for your cluster.

Encryption at rest
☒ **Enable**
Enables encryption of data stored on disk.

Encryption key
The master key that will be used to protect the key used to encrypt data at rest.

☒ **Default key**
An AWS owned key will be used for encryption.

☐ **Customer managed CMK**
Select a customer managed key.

Encryption in transit
☒ **Enable**
Enables encryption of data that moves between the service and client.

Access control
Provides the ability to configure authenticating and authorizing access.

AUTH default user access

AUTH token
The AUTH token used for the cluster.

☐ Show token

At least 16 characters and a maximum of 128 characters, which can be any printable ASCII character except for ' ' (blank space), "" (quotation mark), '/', and '@'.

Mini-hands-on - Securing Redis OSS Cache - ElastiCache

While I was working on the last hands-on lab, I peeked over to the advanced settings. In the Security section, you can enable encryption and rest, configure an encryption master key, enable encryption in transit, and configure access control. I wanted to play around with this.

Hands-on Creating an S3 Bucket

Let's create scalable storage

Create bucket Info

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (Ohio) us-east-2

Bucket type Info

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name Info

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

Format: s3://bucket/prefix

Let's encrypt the storage using server-side encryption with Amazon S3-managed keys (SSE-S3)

Default encryption Info

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type Info

☒ **Server-side encryption with Amazon S3 managed keys (SSE-S3)**

☐ **Server-side encryption with AWS Key Management Service keys (SSE-KMS)**

☐ **Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)**
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

🔔 **Successfully created bucket "jazz-demo-s3-v5"**
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

▶ **Account snapshot - updated every 24 hours** All AWS Regions [View Storage Lens dashboard](#)

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets | Directory buckets

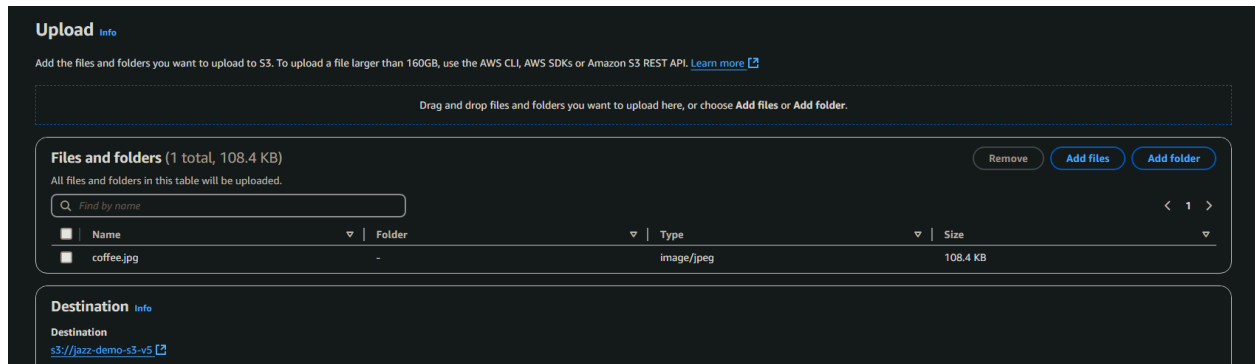
General purpose buckets (1) Info All AWS Regions

Buckets are containers for data stored in S3.

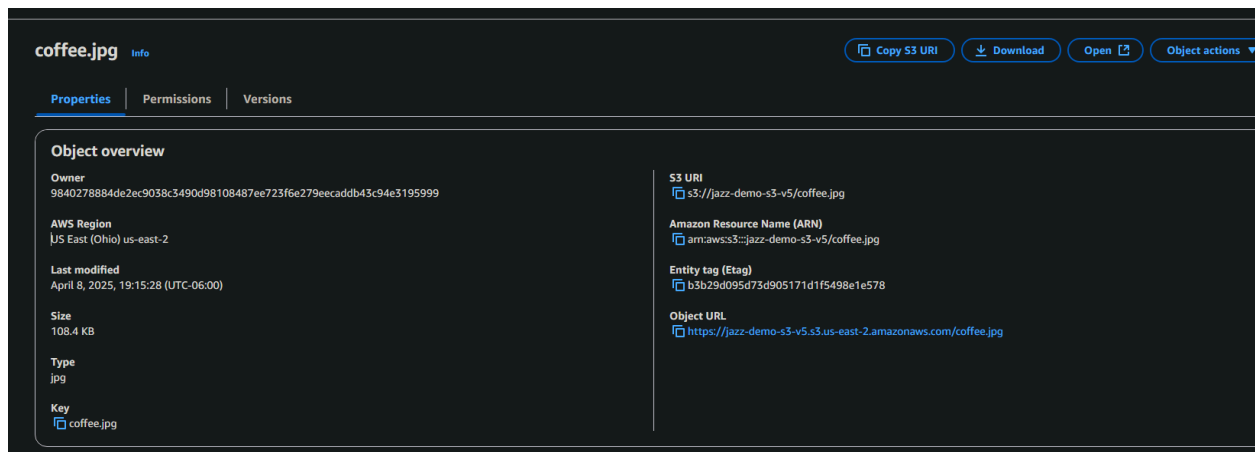
Name	AWS Region	IAM Access Analyzer	Creation date
☒ jazz-demo-s3-v5	US East (Ohio) us-east-2	View analyzer for us-east-2	April 8, 2025, 19:13:21 (UTC-06:00)

Let's start to upload objects to this S3 storage bucket

I'm going to upload coffee.jpg



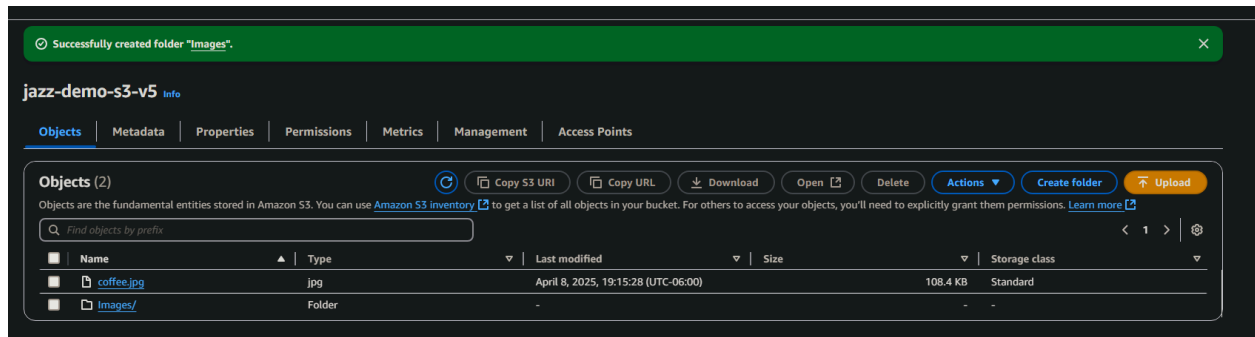
Uploaded



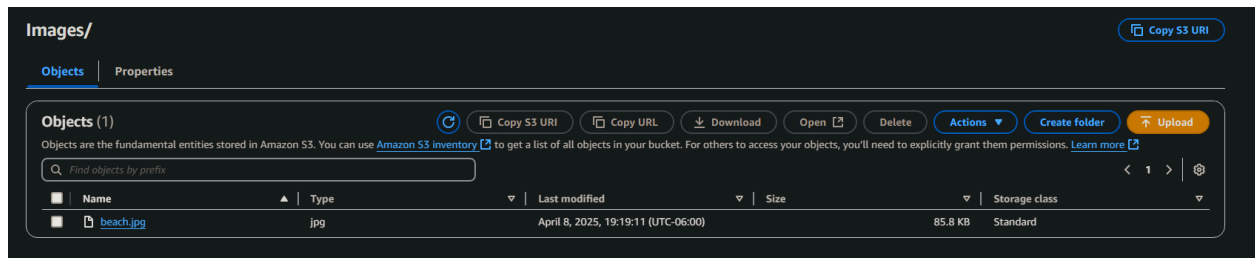
Now, to open up the S3 bucket



Let's create a folder to store jpegs called images



I'm going to upload another jpg called beach.jpg

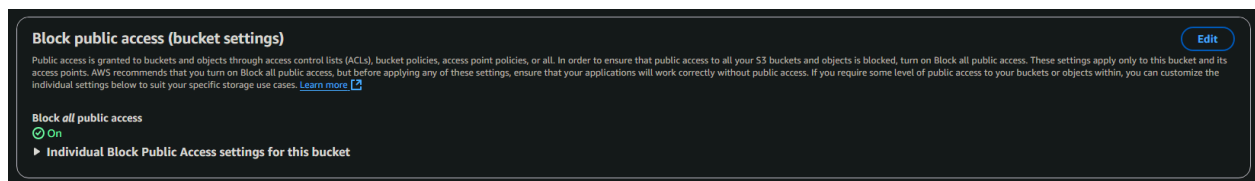




Hands-on Securing S3 Buckets

I noticed when I created my first URL and uploaded coffee.jpg. I could access it directly through Amazon, but not from the public URL. I restricted it ONLY through this way, so I wanted to find a solution to this.

In the permissions, I need to first edit this



Now that's gone, I can work on a new Bucket policy

Let's take a look at the policy generator

A screenshot of the AWS Policy Generator tool. The page has a dark background. At the top left is the Amazon Web Services logo. The main heading is 'AWS Policy Generator'. Below it, a paragraph explains the tool's purpose. The page is divided into three steps. Step 1: 'Select Policy Type' shows a dropdown menu with 'S3 Bucket Policy' selected. Step 2: 'Add Statement(s)' contains several fields: 'Effect' with radio buttons for 'Allow' (selected) and 'Deny'; 'Principal' with a text input field; 'AWS Service' with a dropdown menu showing 'Amazon S3' and a checkbox for 'All Services (***)'; 'Actions' with a dropdown menu showing 'Select Actions --' and a checkbox for 'All Actions (***)'; and 'Amazon Resource Name (ARN)' with a text input field. Below these fields is a link 'Add Conditions (Optional)' and a yellow 'Add Statement' button. Step 3: 'Generate Policy' contains a paragraph explaining that a policy is a document and a link 'Add one or more statements above to generate a policy.' At the bottom of the page, there is a small copyright notice and the 'An amazon.com company' logo.

I want to allow anyone, so I'm changing the Principal to *

With the Actions, I want to select an object.

With the Amazon Resource Name (ARN), I can grab that from my edit bucket policy section. I

also need to add /* at the end

AWS Policy Generator

The AWS Policy Generator is a tool that enables you to create policies that control access to **Amazon Web Services (AWS)** products and resources. For more information about creating policies, see **key concepts in Using AWS Identity and Access Management**. Here are **sample policies**.

Step 1: Select Policy Type

A Policy is a container for permissions. The different types of policies you can create are an IAM Policy, an S3 Bucket Policy, an SNS Topic Policy, a VPC Endpoint Policy, and an SQS Queue Policy.

Select Type of Policy: **S3 Bucket Policy**

Step 2: Add Statement(s)

A statement is the formal description of a single permission. See a **description of elements** that you can use in statements.

Effect: ☒ Allow ☐ Deny

Principal:

Use a comma to separate multiple values.

AWS Service: **Amazon S3** ☐ All Services (**)

Use multiple statements to add permissions for more than one service.

Actions: **-- Select Actions --** ☐ All Actions (**)

Amazon Resource Name (ARN):

ARN should follow the following format: `arn:aws:s3:::{BucketName}/{keyName}`.
Use a comma to separate multiple values.

Add Conditions (Optional)

Add Statement

You added the following statements. Click the button below to Generate a policy.

Principal(s)	Effect	Action	Resource	Conditions
*	Allow	s3:GetObject	arn:aws:s3::jazz-demo-s3-v5/*	None

Step 3: Generate Policy

A policy is a document (written in the **Access Policy Language**) that acts as a container for one or more statements.

Generate Policy **Start Over**

This AWS Policy Generator is provided for informational purposes only; you are still responsible for your use of Amazon Web Services technologies and ensuring that your use is in compliance with all applicable terms and conditions. This AWS Policy Generator is provided as is without warranty of any kind, whether express, implied, or statutory. This AWS Policy Generator does not modify the applicable terms and conditions governing your use of Amazon Web Services technologies.

©2010, Amazon Web Services LLC or its affiliates. All rights reserved.

An Amazon.com company

Looking good

Edit bucket policy [Info](#)

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Policy examples **Policy generator**

Bucket ARN

[arn:aws:s3::jazz-demo-s3-v5](#)

Policy

```
1 {
2   "Id": "Policy1744162985289",
3   "Version": "2012-10-17",
4   "Statement": [
5     {
6       "Sid": "Stmt1744162966750",
7       "Action": [
8         "s3:GetObject"
9       ],
10      "Effect": "Allow",
11      "Resource": "arn:aws:s3::jazz-demo-s3-v5/*",
12      "Principal": "*"
13    }
14  ]
15 }
```

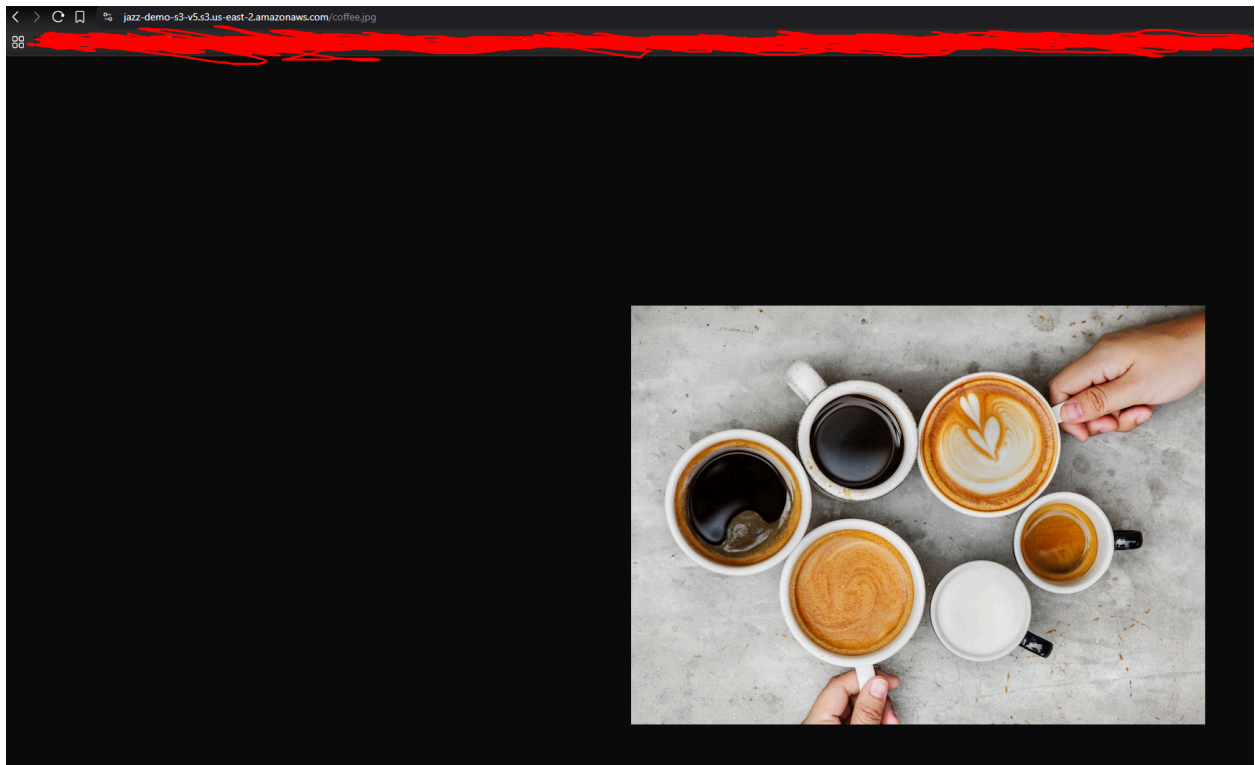
Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

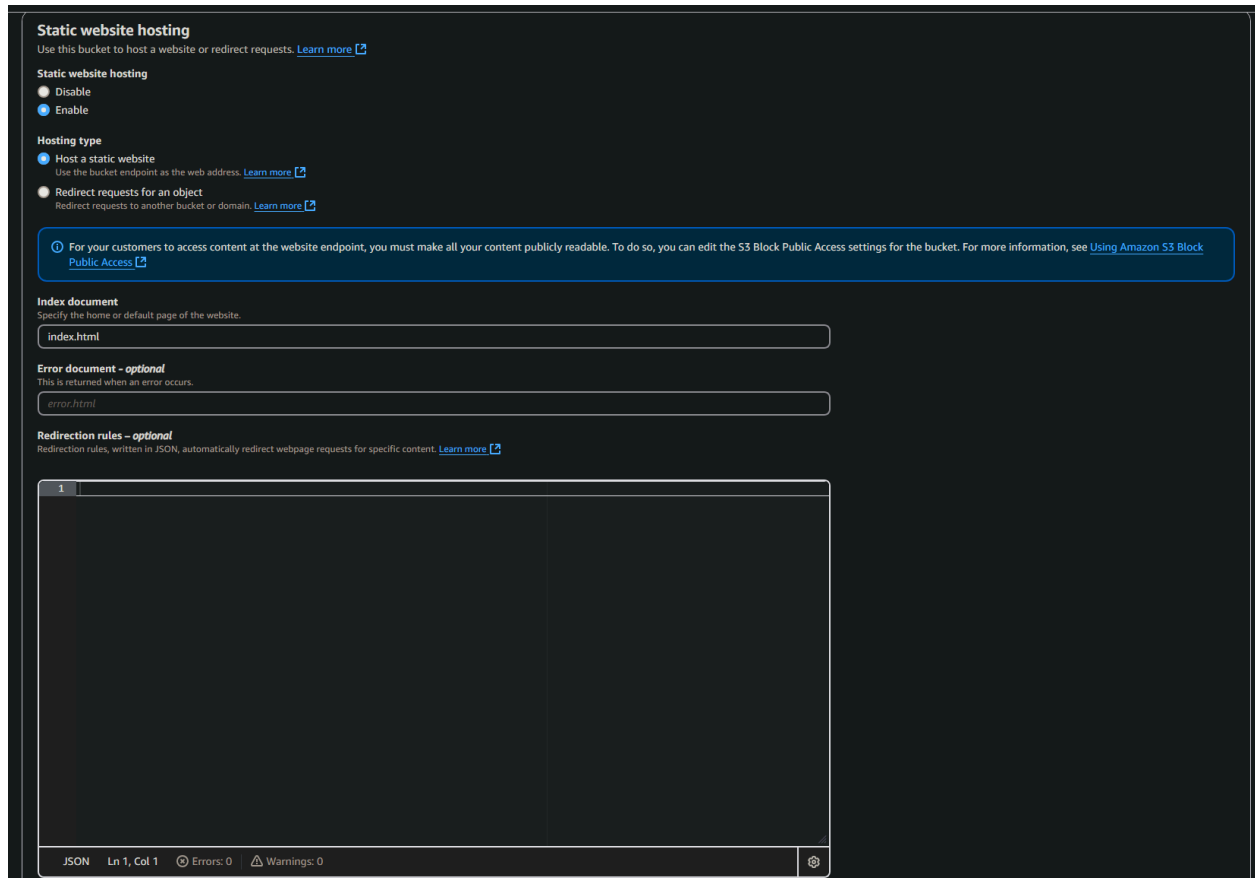
+ Add new statement

Now I can access it through the public URL



Hands-On Creating an S3 Website

First, let me edit the Static Website hosting



The screenshot shows the 'Static website hosting' configuration page in the AWS S3 console. The page has a dark theme. At the top, the title 'Static website hosting' is followed by a subtitle 'Use this bucket to host a website or redirect requests.' and a 'Learn more' link. Below this, the 'Static website hosting' section has two radio buttons: 'Disable' and 'Enable', with 'Enable' selected. The 'Hosting type' section has two radio buttons: 'Host a static website' (selected) and 'Redirect requests for an object'. The 'Host a static website' option has a subtitle 'Use the bucket endpoint as the web address.' and a 'Learn more' link. The 'Redirect requests for an object' option has a subtitle 'Redirect requests to another bucket or domain.' and a 'Learn more' link. A blue information banner states: 'For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see Using Amazon S3 Block Public Access'. Below the banner, the 'Index document' section has a subtitle 'Specify the home or default page of the website.' and a text input field containing 'index.html'. The 'Error document - optional' section has a subtitle 'This is returned when an error occurs.' and a text input field containing 'error.html'. The 'Redirection rules - optional' section has a subtitle 'Redirection rules, written in JSON, automatically redirect webpage requests for specific content.' and a 'Learn more' link. Below this is a large text area for editing JSON rules. At the bottom, a status bar shows 'JSON', 'Ln 1, Col 1', 'Errors: 0', 'Warnings: 0', and a settings icon.

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
☐ Disable
☒ Enable

Hosting type
☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
Specify the home or default page of the website.

Error document - optional
This is returned when an error occurs.

Redirection rules - optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)

1

JSON Ln 1, Col 1 Errors: 0 Warnings: 0

I can now upload a basic homepage with: index.html

Upload

Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (1 total, 597.0 B)

Remove

Add files

Add folder

All files and folders in this table will be uploaded.

Find by name

< 1 >

Name	Folder	Type	Size
index.html	-	text/html	597.0 B

Destination

Info

Destination

s3://jazz-demo-s3-v5

Destination details

Bucket settings that impact new objects stored in the specified destination.

Permissions

Grant public access and access to other AWS accounts.

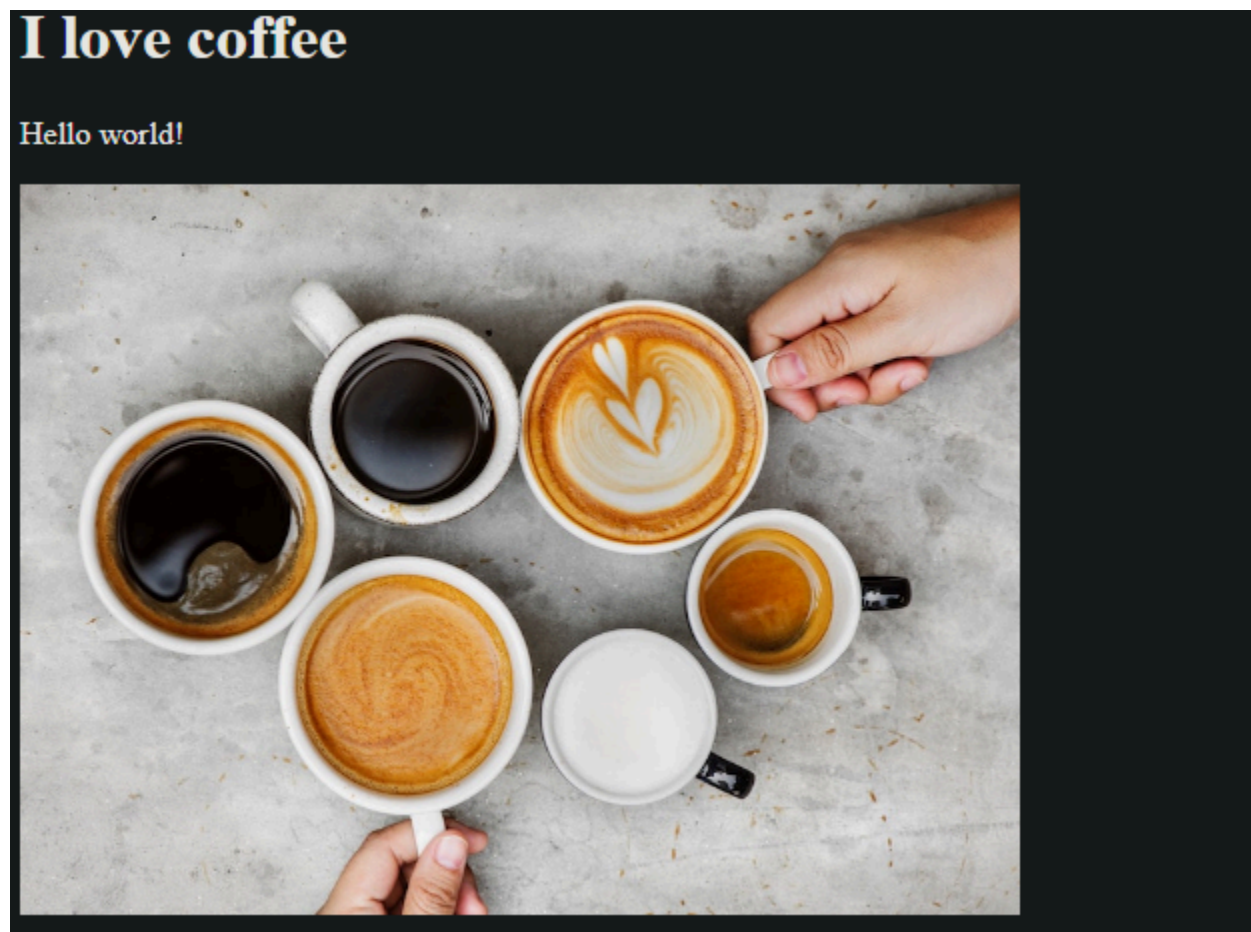
Properties

Specify storage class, encryption settings, tags, and more.

Cancel

Upload

If I go back into Bucket website endpoint, I can access the public ULR now

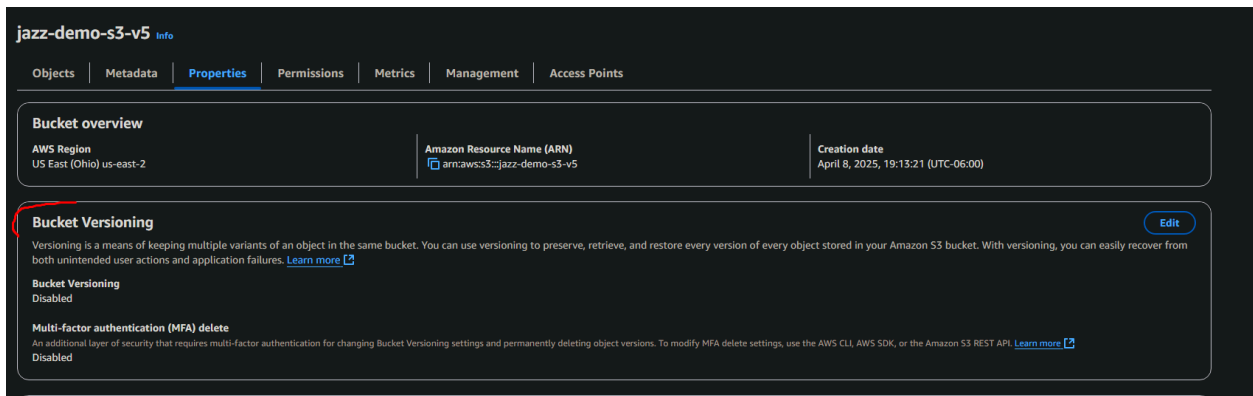


I can also access the beach.jpg



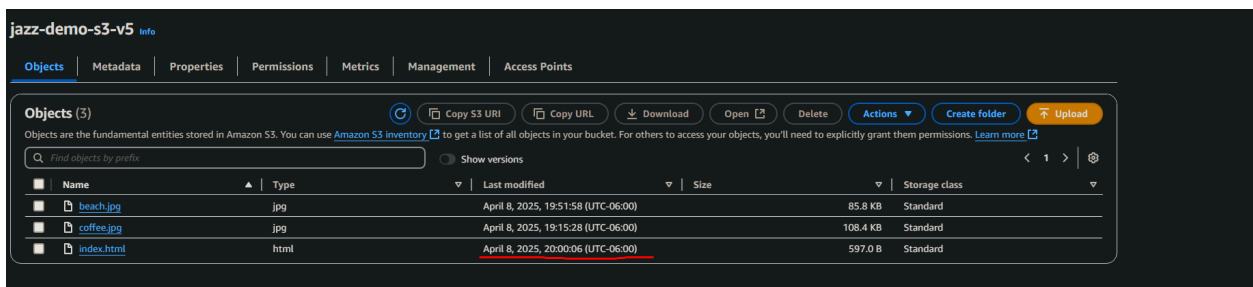
Hands-on S3 Versioning

I can edit bucket versioning in my bucket



I’m going to edit the html file by changing it from I love coffee to I HATE coffee just to see the versioning in effect.

Uploaded the new version



I HATE coffee

Hello world!



I can see the changes over here

Objects (4)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

	Name	Type	Version ID	Last modified	Size	Storage class
☐	beach.jpg	jpg	null	April 8, 2025, 19:51:58 (UTC-06:00)	85.8 KB	Standard
☐	coffee.jpg	jpg	null	April 8, 2025, 19:15:28 (UTC-06:00)	108.4 KB	Standard
☐	index.html	html	1b.mEYEHygGUenm00qD5XvO4Kq-dJtB83	April 8, 2025, 20:00:06 (UTC-06:00)	597.0 B	Standard
☐	index.html	html	null	April 8, 2025, 19:54:23 (UTC-06:00)	597.0 B	Standard

Now, let's say I want to go back to that previous version.

I need to DELETE the second html

Objects (4)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

	Name	Type	Version ID	Last modified	Size	Storage class
☐	beach.jpg	jpg	null	April 8, 2025, 19:51:58 (UTC-06:00)	85.8 KB	Standard
☐	coffee.jpg	jpg	null	April 8, 2025, 19:15:28 (UTC-06:00)	108.4 KB	Standard
☐	index.html	html	1b.mEYEHygGUenm00qD5XvO4Kq-dJtB83	April 8, 2025, 20:00:06 (UTC-06:00)	597.0 B	Standard
☐	index.html	html	null	April 8, 2025, 19:54:23 (UTC-06:00)	597.0 B	Standard

Delete objects Info

- If a folder is selected for deletion, all objects in the folder will be deleted, and any new objects added while the delete action is in progress might also be deleted. If an object is selected for deletion, any new objects with the same name that are uploaded before the delete action is completed will also be deleted.
- Deleting the specified objects can't be undone.

[Learn more](#)

Specified objects

Find objects by name

Name	Version ID	Type	Last modified	Size
index.html	1b.mEYEHygGUenm00qD5XvO4Kq-dJtB83	html	April 8, 2025, 20:00:06 (UTC-06:00)	597.0 B

Permanently delete objects?

To confirm deletion, type *permanently delete* in the text input field.

permanently delete

Cancel

Delete objects

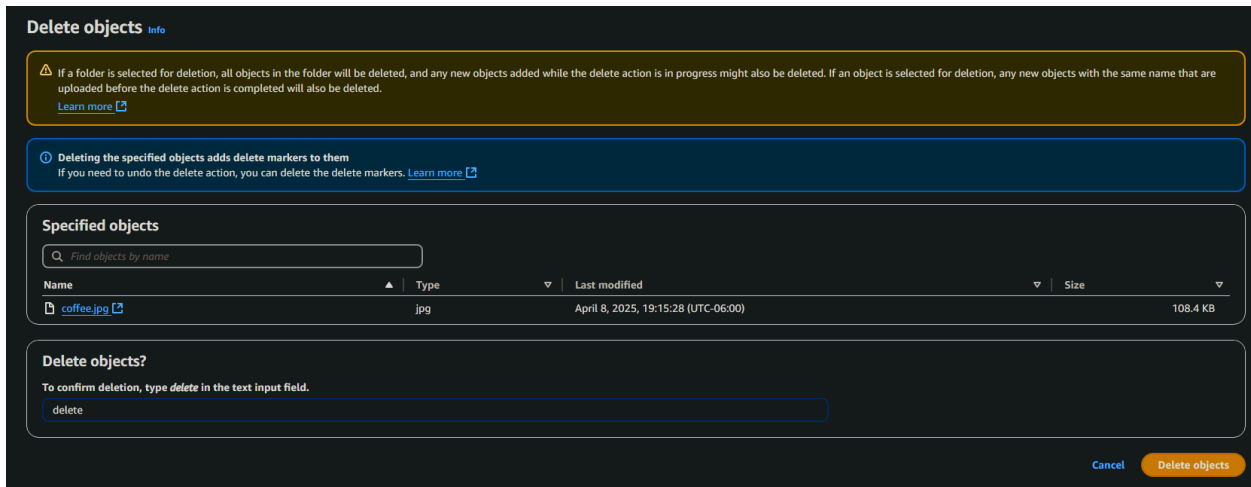
It's now back

I love coffee

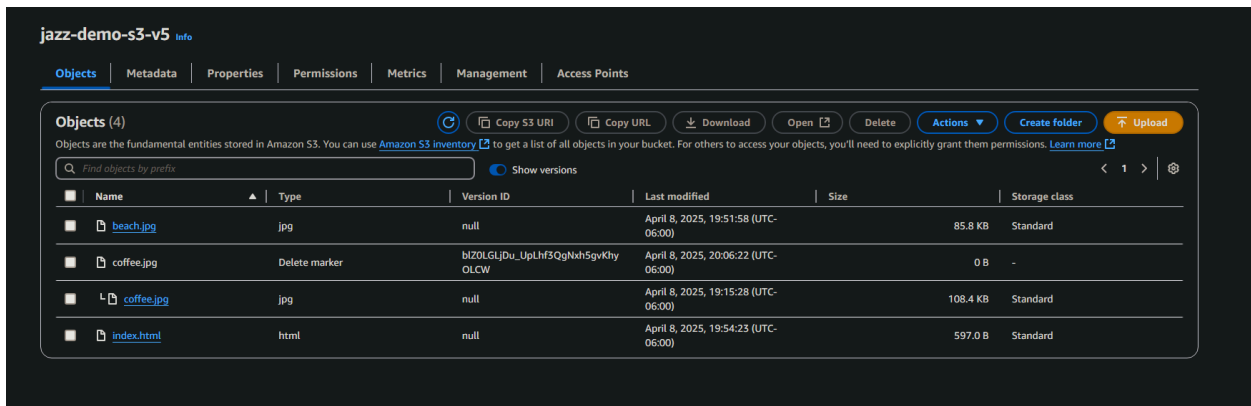
Hello world!



Now, I’m going to delete coffee.jpg. This will create a delete marker and it doesn’t actually delete the underlying object



It still hows when I toggle ‘Show versions’



This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>CQD9FMP19N0Q8TBZ</RequestId>
  <HostId>C1DsLY8nGy+61i2Ez7dEdbd3XwchYTo2AF3b83kGwiSF8XeBZdAkIbKSopWTA7ETqwZwy0zIYqU=</HostId>
</Error>
```

To fix this, I'm going to delete the marker. It's back now

I love coffee

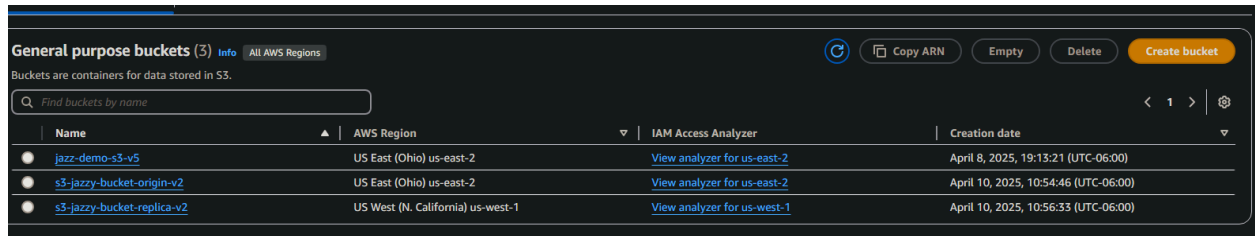
Hello world!



Hands-on S3 Replication

I want to check out replication between two different S3 buckets in different AZs

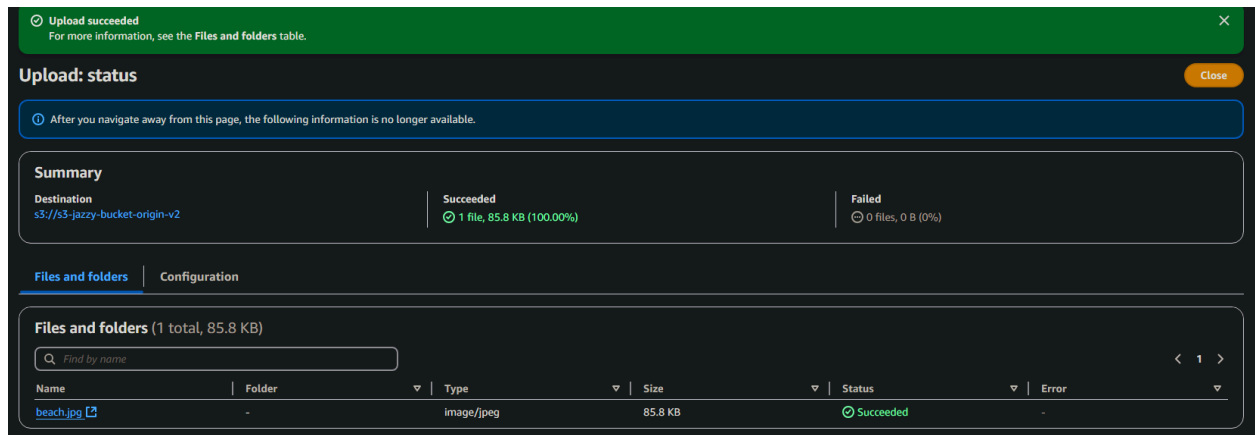
s3-jazzy-bucket-origin-v2 is in us-east-2, s3-jazzy-bucket-replica-v2 is in us-west-1



The screenshot shows the AWS S3 console's 'General purpose buckets' page. It lists three buckets: 'jazz-demo-s3-v5', 's3-jazzy-bucket-origin-v2', and 's3-jazzy-bucket-replica-v2'. The first two are in the 'us-east-2' region, and the third is in the 'us-west-1' region. Each bucket has a link to 'View analyzer'.

Name	AWS Region	IAM Access Analyzer	Creation date
jazz-demo-s3-v5	US East (Ohio) us-east-2	View analyzer for us-east-2	April 8, 2025, 19:13:21 (UTC-06:00)
s3-jazzy-bucket-origin-v2	US East (Ohio) us-east-2	View analyzer for us-east-2	April 10, 2025, 10:54:46 (UTC-06:00)
s3-jazzy-bucket-replica-v2	US West (N. California) us-west-1	View analyzer for us-west-1	April 10, 2025, 10:56:33 (UTC-06:00)

Now, I'm going to upload an image to the origin bucket. Let's do the beach.jpg



The screenshot shows the 'Upload: status' page in the AWS S3 console. It displays a summary of the upload process, indicating that the file 'beach.jpg' (85.8 KB) was successfully uploaded to the destination 's3://s3-jazzy-bucket-origin-v2'. The 'Files and folders' section shows a table with one entry: 'beach.jpg' with a status of 'Succeeded'.

Upload: status

After you navigate away from this page, the following information is no longer available.

Summary

Destination	Succeeded	Failed
s3://s3-jazzy-bucket-origin-v2	1 file, 85.8 KB (100.00%)	0 files, 0 B (0%)

Files and folders (1 total, 85.8 KB)

Name	Folder	Type	Size	Status	Error
beach.jpg	-	image/jpeg	85.8 KB	Succeeded	-

Now, with the origin bucket, I need to make a new replication rule in the management tab

Create replication rule [Info](#)

Replication rule configuration

Replication rule name

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status
Choose whether the rule will be enabled or disabled when created.
☒ Enabled
☐ Disabled

Priority
The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.

Source bucket

Source bucket name
s3-jazzy-bucket-origin-v2

Source Region
US East (Ohio) us-east-2

Choose a rule scope
☐ Limit the scope of this rule using one or more filters
☒ Apply to all objects in the bucket

For the destination, I need to pick the replication bucket: s3-jazzy-bucket-replica-v2

Destination

Destination
You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. [Learn more](#) or see [Amazon S3 pricing](#)

☒ Choose a bucket in this account
☐ Specify a bucket in another account

Bucket name
Choose the bucket that will receive replicated objects.
 [Browse S3](#)

Destination Region
US West (N. California) us-west-1

Done

Replication configuration successfully updated
If changes to the configuration aren't displayed, choose the refresh button. Changes apply only to new objects. To replicate existing objects with this configuration, choose **Create replication job**.

Replication rules

Replication enables automatic and asynchronous copying of objects across buckets in the same or different AWS Regions. A replication configuration is a set of rules that define what options should be applied to a group of objects during replication.

Replication configuration settings

Configuration settings affect all replication rules in the bucket.

Source bucket
s3-jazzy-bucket-origin-v2

Source Region
US East (Ohio) us-east-2

IAM role
s3cnr_role_for_s3-jazzy-bucket-origin-v2

Create replication job

Edit

Replication rules (1)

Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioning replicas to another storage class, and more. [Learn more](#)

Replication rule name	Status	Destination bucket	Destination Region	Priority	Scope	Storage class	Replica owner	Replication Time Control	KMS-encrypted objects (SSE-KMS or DSSE-KMS)	Replica modification sync
DemoReplication Rule	Enabled	s3://s3-jazzy-bucket-replica-v2	US West (N. California) us-west-1	0	Entire bucket	Same as source	Same as source	Disabled	Do not replicate	Disabled

Now, I uploaded coffee.jpg into the origin bucket

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

Name	Type	Last modified	Size	Storage class
beach.jpg	jpg	April 10, 2025, 11:11:54 (UTC-06:00)	85.8 KB	Standard
coffee.jpg	jpg	April 10, 2025, 11:10:18 (UTC-06:00)	108.4 KB	Standard

Both images are replicated to the replica s3 bucket

s3-jazzy-bucket-replica-v2

Objects

Properties

Permissions

Metrics

Management

Access Points

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

Name	Type	Last modified	Size	Storage class
beach.jpg	jpg	April 10, 2025, 11:11:54 (UTC-06:00)	85.8 KB	Standard
coffee.jpg	jpg	April 10, 2025, 11:10:18 (UTC-06:00)	108.4 KB	Standard

If I show the version with both the replica and the origin, the version IDs match

s3-jazzy-bucket-replica-v2

Info

Objects

Properties

Permissions

Metrics

Management

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

< 1 >

	Name	Type	Version ID	Last modified	Size	Storage class
	beach.jpg	jpg	VnpsRdb_ITz6m508AAy8Vd8_0PTo_Xf5	April 10, 2025, 11:11:54 (UTC-06:00)	85.8 KB	Standard
	coffee.jpg	jpg	IcwkUr9E_JEDPDGn8dtatLrFCmhAjKZ	April 10, 2025, 11:10:18 (UTC-06:00)	108.4 KB	Standard

s3-jazzy-bucket-origin-v2

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

< 1 >

	Name	Type	Version ID	Last modified	Size	Storage class
	beach.jpg	jpg	VnpsRdb_ITz6m508AAy8Vd8_0PTo_Xf5	April 10, 2025, 11:11:54 (UTC-06:00)	85.8 KB	Standard
	coffee.jpg	jpg	IcwkUr9E_JEDPDGn8dtatLrFCmhAjKZ	April 10, 2025, 11:10:18 (UTC-06:00)	108.4 KB	Standard

Now, I’m going to upload beach.jpg again to the origin bucket

Upload succeeded

For more information, see the Files and folders table.

Close

Upload: status

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://s3-jazzy-bucket-origin-v2

Succeeded

1 file, 85.8 KB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 85.8 KB)

Find by name

< 1 >

Name	Folder	Type	Size	Status	Error
beach.jpg	-	image/jpeg	85.8 KB	Succeeded	-

There are now two different versions

s3-jazzy-bucket-origin-v2

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (3)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

< 1 >

	Name	Type	Version ID	Last modified	Size	Storage class
	beach.jpg	jpg	ppfBayM4uyCehYM8qxZ7I5qYCLlt4kVy	April 10, 2025, 11:14:42 (UTC-06:00)	85.8 KB	Standard
	beach.jpg	jpg	VnpsRdb_ITz6m50BAAY8Vd8_0PTo_Xf5	April 10, 2025, 11:11:54 (UTC-06:00)	85.8 KB	Standard
	coffee.jpg	jpg	IcwxkUr9E_JEDPDGn8dtatLrFCmhAjkZ	April 10, 2025, 11:10:18 (UTC-06:00)	108.4 KB	Standard

Same with the replica

s3-jazzy-bucket-replica-v2

Objects

Properties

Permissions

Metrics

Management

Access Points

Objects (3)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

< 1 >

	Name	Type	Version ID	Last modified	Size	Storage class
	beach.jpg	jpg	ppfBayM4uyCehYM8qxZ7I5qYCLlt4kVy	April 10, 2025, 11:14:42 (UTC-06:00)	85.8 KB	Standard
	beach.jpg	jpg	VnpsRdb_ITz6m50BAAY8Vd8_0PTo_Xf5	April 10, 2025, 11:11:54 (UTC-06:00)	85.8 KB	Standard
	coffee.jpg	jpg	IcwxkUr9E_JEDPDGn8dtatLrFCmhAjkZ	April 10, 2025, 11:10:18 (UTC-06:00)	108.4 KB	Standard

Now, I will go back to my S3 management tab to edit the replication options to enable delete marker replication

Additional replication options

Replication Time Control (RTC)

Replication Time Control replicates 99.99% of new objects within 15 minutes and includes replication metrics. Additional fees will apply. [Learn more](#)

Replication metrics

With replication metrics, you can monitor the total number and size of objects that are pending replication, and the maximum replication time to the destination Region. You can also view and diagnose replication failures. CloudWatch metrics fees apply. [Learn more](#) or see [Amazon CloudWatch pricing](#)

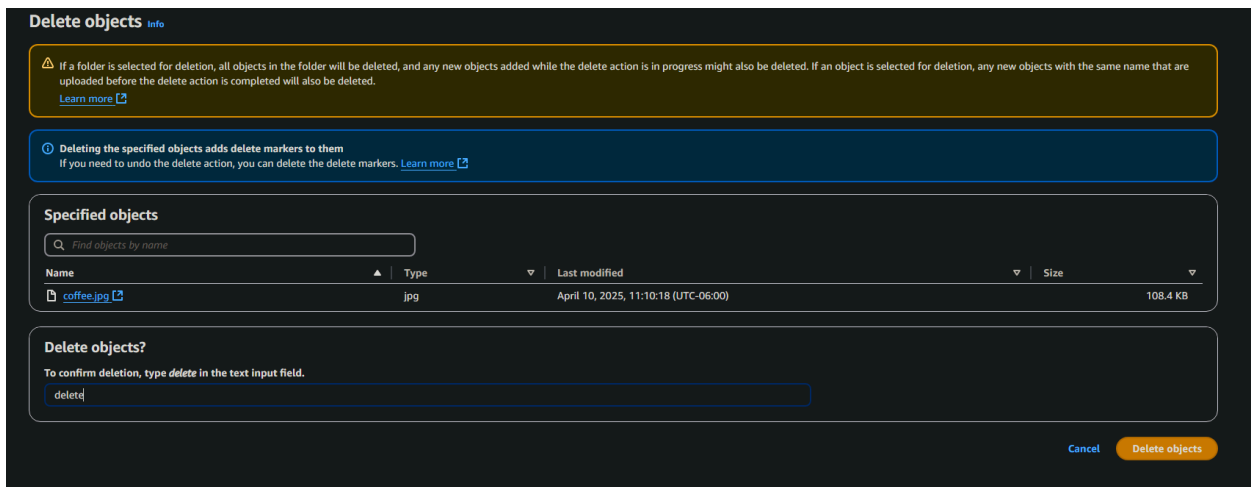
☒ Delete marker replication

Delete markers created by S3 delete operations will be replicated. Delete markers created by lifecycle rules are not replicated. [Learn more](#)

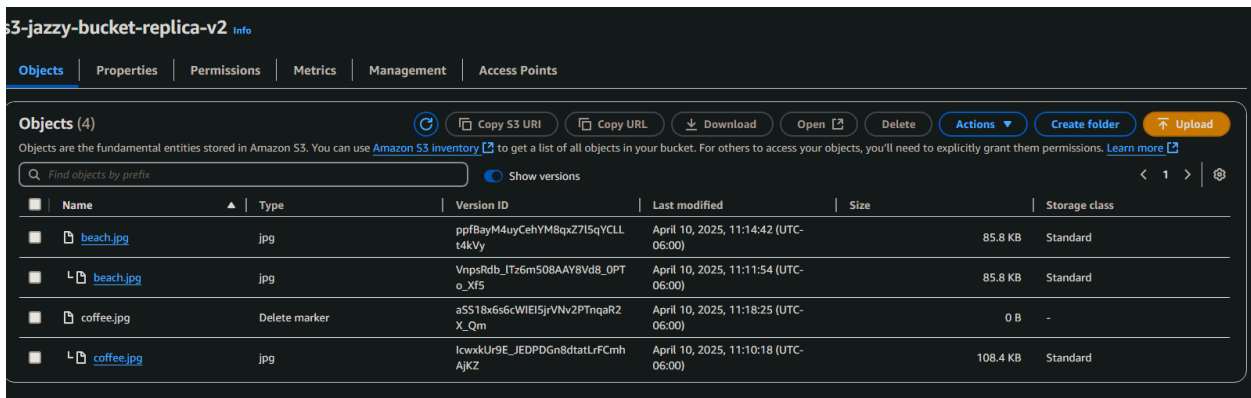
☐ Replica modification sync

Replicate metadata changes made to replicas from the destination bucket to the source bucket. [Learn more](#)

So now, if I go to the origin bucket and delete coffee.jpg, look at this



In the replica, I can see the delete marker here



Hands-on S3 Event Notifications

Let's create a bucket first

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (Ohio) us-east-2

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

demo-jazzy-v3-event-notifications

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

Now I'm creating an event notification

Create event notification [Info](#)

To enable notifications, you must first add a notification configuration that identifies the events you want Amazon S3 to publish and the destinations where you want Amazon S3 to send the notifications.

General configuration

Event name

DemoEventNotification

Event name can contain up to 255 characters.

Prefix - optional
Limit the notifications to objects with key starting with specified characters.

images/

Suffix - optional
Limit the notifications to objects with key ending with specified characters.

.jpg

Event types

Specify at least one event for which you want to receive notifications. For each group, you can choose an event type for all events, or you can choose one or more individual events.

Object creation

☒ **All object create events**
s3:ObjectCreated:*

☐ **Put**
s3:ObjectCreated:Put

☐ **Post**
s3:ObjectCreated:Post

☐ **Copy**
s3:ObjectCreated:Copy

☐ **Multipart upload completed**
s3:ObjectCreated:CompleteMultipartUpload

I need to create an SQS first real quick

Create queue

Details

Type
Choose the queue type for your application or cloud infrastructure.

☒ **Standard** [Info](#)
At-least-once delivery, message ordering isn't preserved

- At-least once delivery
- Best-effort ordering

☐ **FIFO** [Info](#)
First-in-first-out delivery, message ordering is preserved

- First-in-first-out delivery
- Exactly-once processing

You can't change the queue type after you create a queue.

Name

DemoS3Notification

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

I tried to put this in and I got this error

SQS queue

DemoS3Notification

Unknown Error
An unexpected error occurred. Try again later. If the error persists, contact [AWS Support for assistance](#).

[Diagnose with Amazon Q](#)

▼ API response

Unable to validate the following destination configurations

Cancel Save changes

It looks like I need to adjust the Queue policies of my SQS



Let's take a look at the Policy Generator. I need to allow anyone to write to the SQS queue. (I would never do this outside of a lab.)

I can now save this new access policy



Send and receive messages

Send messages to and receive messages from a queue.

Send message Info

Clear content

Send message

Message body
Enter the message to send to the queue.

Enter message

Delivery delay Info

0

Seconds

Should be between 0 seconds and 15 minutes.

► **Message attributes - Optional** Info

Receive messages Info

Edit poll settings

Stop polling

⌵ Poll for messages

Messages available

0

Polling duration

30

Maximum message count

10

Polling progress

1 receives/second

50%

Messages (1)

Q Search messages

View details

Delete

■	ID	▼	Sent	▲	Size	▼	Receive count	▼
■	4a9f87dc-69f4-47db-8149-5378285e371e		2025-04-14T20:49:06:00		244 bytes		1	

Send message Info

Message body
Enter the message to send to the queue.

Message: a08c4355-1437-4ecc-a3d6-738c71d300a6

Body
Attributes
Details

```
[{"Records":[{"eventVersion":"2.1","eventSource":"aws:s3","awsRegion":"us-east-2","eventTime":"2025-04-15T02:50:51.806Z","eventName":"ObjectCreated:Put","userIdentity":{"principalId":"AWS:AIDAGELKNWTJQFA3KJS6Y"},"requestParameters":{"sourceAddress":"38.70.249.11"},"responseElements":{"x-amz-request-id":"WWW5AGV03GHSDERP4","x-amz-id-2":"99JCWVEYvOPMumfOGAidRfrE71B0Zrc7vM57nzbcJJBAICQkPoplBPOTnzwrkKV1UW7SRadGB2Xz2Q1ogva+gSVMB+FSY8BMhp-phs=1,k3","x-s3-changed-version":["D"],"configuration":{"demo-jazzy-v3-event-notifications":{"bucket":{"name":"demo-jazzy-v3-event-notifications"},"ownerIdentity":{"principalId":"A3CRM2LCPAENR"},"arn":"arn:aws:s3:::demo-jazzy-v3-event-notifications"},"object":{"key":"coffee.jpg","size":110985,"eTag":"b3b29d095d73d905171d1f5498e1e578","sequencer":"0067FDC98BC484B7F7"}}}]}
```

Receive messages Info

Messages available
0

Messages (2)

ID	Sent	Size	Receive count
a49f87dc-69f4-47db-8149-5378285e371e	2025-04-14T20:49-06:00	244 bytes	3
a08c4355-1437-4ecc-a3d6-738c71d300a6	2025-04-14T20:50-06:00	813 bytes	1

Hands-on S3 Encryption

First, let's create an S3 bucket

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (Ohio) us-east-2

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

demo-encryption-jazzy-v2

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

☒ Choose bucket

Format: s3://bucket/prefix

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Disable

☒ Enable

Let's have the encryption standard be server-side encryption with Amazon S3 managed keys (SSE-S3)

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

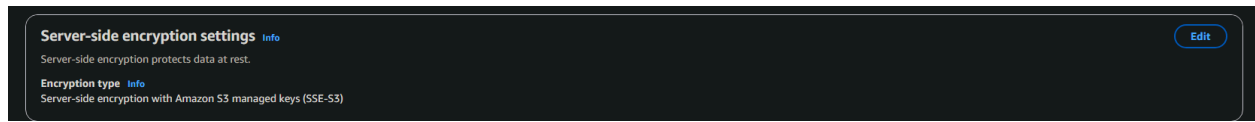
☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

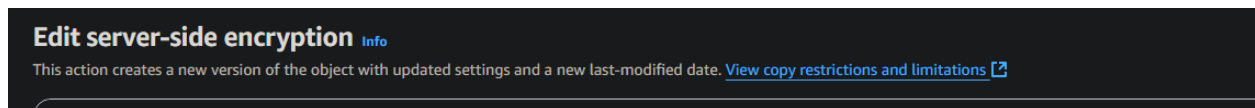
☐ Disable

☒ Enable

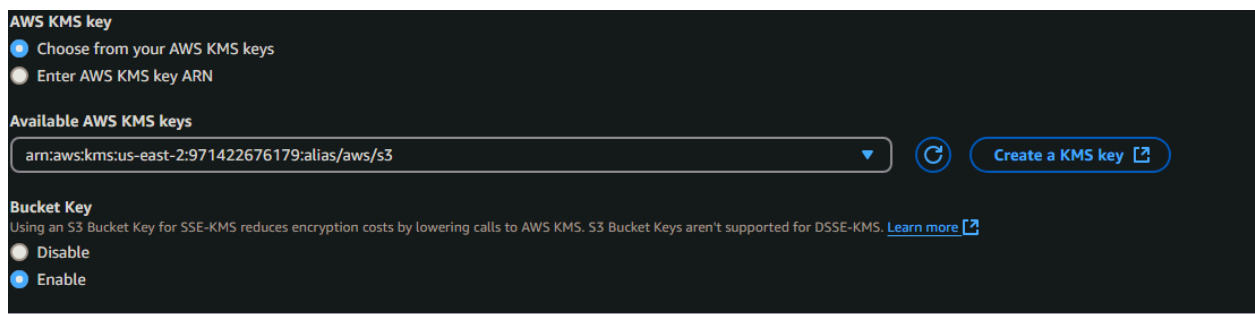
To verify encryption, I will upload a photo to the bucket - coffe.jpg



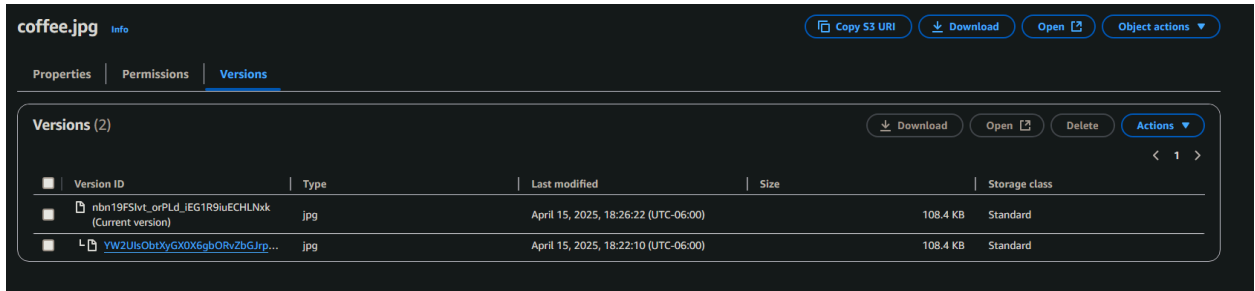
If I try to edit the photo, it will create a new version of the object



I can change the encryption type to server-side encryption with AWS Key Management Service keys (SSE-KMS). I can also choose the default key



Now I have two versions of coffee.jpg



coffee.jpg [Info](#)

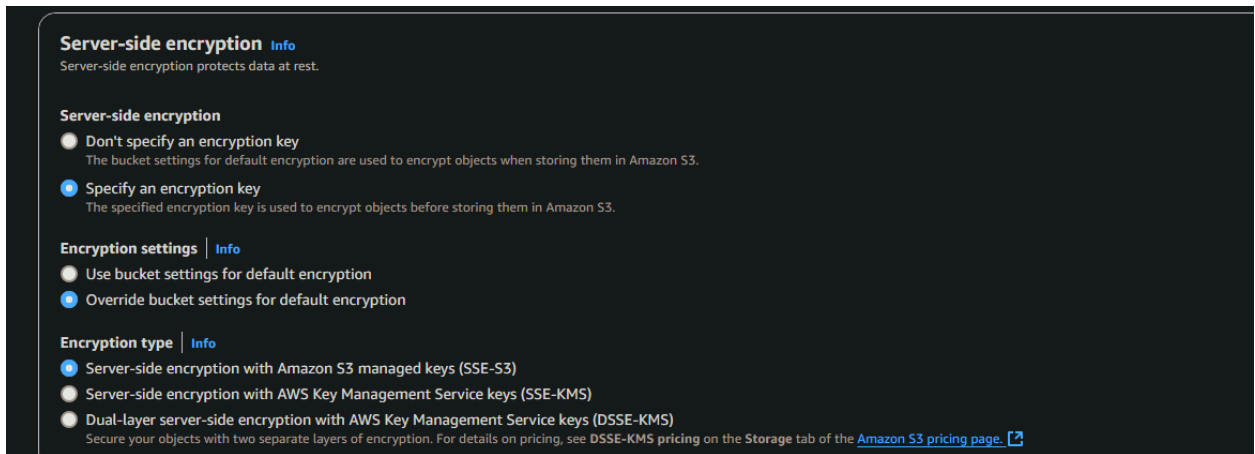
[Copy S3 URI](#) [Download](#) [Open](#) [Object actions](#)

Properties | Permissions | **Versions**

Versions (2) [Download](#) [Open](#) [Delete](#) [Actions](#)

Version ID	Type	Last modified	Size	Storage class
nbn19FSlt_orPLd_IeG1R9IuECHLNsk (Current version)	jpg	April 15, 2025, 18:26:22 (UTC-06:00)	108.4 KB	Standard
YW2UlsObtXyGX0X6gbORvZbGJrp...	jpg	April 15, 2025, 18:22:10 (UTC-06:00)	108.4 KB	Standard

Let's repeat it. I'm going to add beach.jpg. I can also enable encryption when I upload the jpg with the correct encryption type.



Server-side encryption [Info](#)
Server-side encryption protects data at rest.

Server-side encryption

- ☐ Don't specify an encryption key
The bucket settings for default encryption are used to encrypt objects when storing them in Amazon S3.
- ☒ Specify an encryption key
The specified encryption key is used to encrypt objects before storing them in Amazon S3.

Encryption settings | [Info](#)

- ☐ Use bucket settings for default encryption
- ☒ Override bucket settings for default encryption

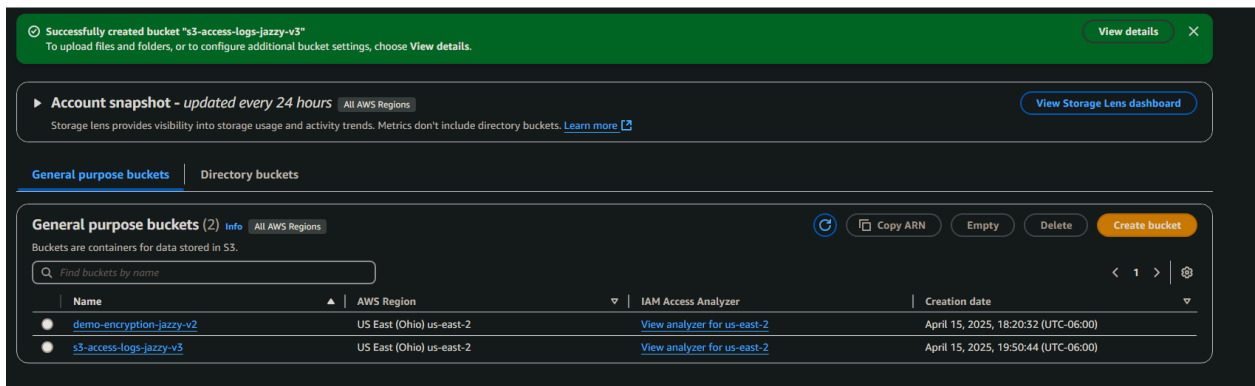
Encryption type | [Info](#)

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

Hands-on S3 Access Logs

I want to audit all access to S3 buckets. Any request made to an S3 from any account, authorized or denied, will be logged into another S3 bucket.

Let's create a bucket. This is the logging bucket.



Successfully created bucket "s3-access-logs-jazzy-v3"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

► Account snapshot - updated every 24 hours [All AWS Regions](#) [View Storage Lens dashboard](#)
Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets | Directory buckets

General purpose buckets (2) [Info](#) [All AWS Regions](#) [Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	IAM Access Analyzer	Creation date
demo-encryption-jazzy-v2	US East (Ohio) us-east-2	View analyzer for us-east-2	April 15, 2025, 18:20:32 (UTC-06:00)
s3-access-logs-jazzy-v3	US East (Ohio) us-east-2	View analyzer for us-east-2	April 15, 2025, 19:50:44 (UTC-06:00)

I'm going to use a previous bucket, demo-encryption-jazzy-v2. I need to edit the server access logging.

Edit server access logging

Server access logging

Log requests for access to your bucket. [Learn more](#)

Server access logging

☐ Disable

☒ Enable

⚠ Bucket policy will be updated

When you enable server access logging, the S3 console automatically updates your bucket policy to include access to the S3 log delivery group.

Destination

Specify a destination bucket in the US East (Ohio) us-east-2 Region. To store your logs under a particular prefix, make sure that you include a slash (/) after the name of the prefix. Otherwise, the prefix will be added to the name of your log files.

s3://s3-access-logs-jazzy-v3

Browse S3

Format: s3://<bucket>/<optional-prefix-with-path>

Destination Region

US East (Ohio) us-east-2

Destination bucket name

s3-access-logs-jazzy-v3

Destination prefix

-

Log object key format

☒ [DestinationPrefix][YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]

☐ [DestinationPrefix][SourceAccountId]/[SourceRegion]/[SourceBucket]/[YYYY]/[MM]/[DD]/[YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]

To speed up analytics and query applications, use this format.

Log object key example

2025-07-01-10-12-56-[UniqueString]

Cancel

Save changes

I also gave it the destination of the access logs s3 bucket.

Let's generate some traffic.

Hands-on Pre-Signed URLs

I want to have a pre-signed URL that will allow users to inherit the permissions of the user that generates the URL for GET / Put. It carries over credentials. I can send the URL to the target user.

So, with a previous bucket: demo-encryption-jazzy-v2, when I try to open up ai.jpg from the object URL, I get this

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
▼ <Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>K9TV4HW1Q1PCB1RM</RequestId>
  <HostId>+ZHTJ8dq21nkEHos7uqY3azTRjjWrcrPhuYbH6a1zJVnZq5bv+J08kFeinNF4zaSfulD590P7G0=</HostId>
</Error>
```

To mitigate this, now, I can simply select Object actions and select *Share with a pre-signed URL* with ai.jpg. Let's have it be available for the next 5 minutes

Share "ai.jpg" with a presigned URL



Presigned URLs are used to grant access to an object for a limited time. [Learn more](#) 

 Anyone can access the object with this presigned URL until it expires, even if the bucket, and object are private.

Time interval until the presigned URL expires

Using the S3 console, you can share an object with a presigned URL for up to 12 hours or until your session expires. To create a presigned URL with a longer time interval, use the AWS CLI or AWS SDK. Time intervals for presigned URLs can be restricted by your IAM policy.

- ☒ Minutes
☐ Hours

Number of minutes

10



Must be a whole number between 1 and 720.

After you create the presigned URL, it's automatically copied to your clipboard.

Cancel

Create presigned URL

Now, in a private window, I can use this [link](#) (Which you will not be able to use because of the expiration)

